

Bayesian Quantile Deep Echo State Networks

Antonio Aguirre¹

Raquel Prado¹

Bruno Sansó¹

¹Department of Statistics, University of California, Santa Cruz

July 8, 2026

Abstract

Conditional quantiles provide distributional forecast summaries for asymmetric losses, tail-risk assessment, and prediction intervals. We develop the quantile deep echo state network (Q-DESN), a Bayesian quantile readout for fixed deep echo state network features. Conditional on reservoir construction, scaling, and washout, the DESN states define a deterministic high-dimensional design matrix, and posterior inference is restricted to the readout model. A quantile-fixed extended asymmetric Laplace (exAL) working likelihood places the target quantile at a regularized linear readout. Ridge priors provide dense regularization, while the regularized horseshoe gives an adaptive shrinkage alternative with closed-form scale updates under the representation used here. Posterior computation is carried out with a Markov chain Monte Carlo (MCMC) sampler and a model-specific variational Bayes approximation that uses a Laplace-Delta treatment of the non-conjugate scale-asymmetry block. For multi-quantile reporting, we distinguish independent single-level fits with post hoc monotone synthesis from a joint quantile-vector readout that places a regularized-horseshoe prior on adjacent coefficient innovations across the quantile grid. This joint prior gives a soft non-crossing mechanism: it shrinks quantile-indexed readout changes toward coherent paths, while exact monotonicity is enforced only by an additional rearrangement or projection step. Simulation studies evaluate single-quantile recovery, held-out forecast accuracy, runtime, and joint multi-quantile coherence under pinned synthetic protocols. GloFAS streamflow and PriceFM electricity-price applications illustrate audited forecast-calibration and benchmark-comparison workflows.

1 Introduction

Forecast evaluation often requires summaries beyond the conditional mean. Conditional means are natural under squared-error loss, but asymmetric decision losses, tail-risk assessment, and interval forecasts call for conditional quantiles. Under asymmetric piecewise-linear loss, conditional quantiles are optimal point forecasts, and pairs of quantiles define interval forecasts (Koenker and Bassett, 1978; Koenker, 2005; Gneiting, 2011). Let y_{t+h} denote the future response at horizon h , and let $\mathcal{F}_t$ be the sigma-field generated by the information available at forecast origin t , including past responses and observed or scenario-specified covariates. With $F_{t,h}(u) = \Pr(y_{t+h} \leq u \mid \mathcal{F}_t)$, define

the level- p_0 conditional quantile by the generalized inverse

$$Q_{t,h}(p_0) = F_{t,h}^{-1}(p_0) = \inf\{u : F_{t,h}(u) \geq p_0\}, \quad p_0 \in (0, 1).$$

A single quantile level is sufficient when the decision target is fixed; a grid of fitted levels can be post-processed into an estimated predictive quantile function for interval construction or distributional forecast evaluation (Gneiting and Raftery, 2007; Gneiting and Katzfuss, 2014; Barlow and Brunk, 1972; Chernozhukov et al., 2010).

For nonlinear dynamic data, quantile forecast accuracy also depends on how temporal dependence is represented. Fixed reservoir features provide one computationally convenient representation: they encode recent history while leaving the recurrent weights outside the posterior inference problem. In the standard echo state network (ESN) construction used here, input and recurrent reservoir weights are generated once and then held fixed, so that only a readout is estimated (Jaeger, 2001; Lukoševičius and Jaeger, 2009; Lukoševičius, 2012). A deep echo state network (DESN) extends this idea by stacking reservoir layers. Conditional on the selected architecture, random seed, scaling constants, reducers, and washout convention, the resulting reservoir states define a deterministic nonlinear feature matrix. Bayesian uncertainty in this article is therefore conditional on the constructed reservoir and is concentrated in the readout model.

The quantile deep echo state network (Q-DESN) combines this fixed DESN design with a Bayesian quantile readout. For a chosen level p_0 , the readout location $\mu_t = \mathbf{x}_t^\top \boldsymbol{\beta}$ is linked to the response through the quantile-fixed generalized asymmetric Laplace (GAL) distribution of Yan et al. (2025). Throughout this article, exAL denotes this quantile-fixed GAL working likelihood. Under this parameterization, μ_t is the model’s p_0 th conditional quantile, while the exAL asymmetry parameter allows mode, skewness, and tail behavior to vary relative to the asymmetric Laplace special case.

The exAL likelihood is used as a working likelihood in the Bayesian quantile-regression sense. It defines a posterior target for quantile-specific readout inference without asserting that exAL is the data-generating error law (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). Posterior intervals for readout quantities and forecast bands generated from the fitted likelihood are therefore model-based summaries conditional on the fixed reservoir and the working likelihood. Coverage is treated as an empirical diagnostic, assessed through simulation or validation data rather than assumed from the posterior alone.

High-dimensional readouts require regularization. The default coefficient prior is ridge, which gives dense Gaussian regularization. As an adaptive alternative, the regularized horseshoe introduces global-local shrinkage and a slab component for large coefficients (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, the local, global, and slab scale updates retain closed-form inverse-gamma forms (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). Posterior computation includes an MCMC sampler for the Q-DESN hierarchy. A mean-field variational Bayes approximation uses Laplace and Delta-method approximations for the non-conjugate exAL scale and asymmetry block. We refer to this implementation as VB-LD when

the specific Laplace–Delta approximation needs to be distinguished. Its role is computational: it approximates the Q–DESN posterior under the working likelihood and is not presented as a general claim about variational-inference methodology.

The paper makes five contributions. First, Q–DESN formulates Bayesian quantile readout inference conditional on fixed deep ESN features. Second, the readout combines a quantile-fixed exAL working likelihood with the default ridge prior and a regularized horseshoe adaptive-shrinkage alternative. Third, model-specific MCMC and variational Bayes updates are derived for the resulting hierarchy. Fourth, multi-quantile reporting is treated through two distinct routes: independent single-level fits followed by isotonic or rearrangement synthesis, and a joint quantile-vector readout with regularized-horseshoe shrinkage across adjacent quantile-specific slopes. Fifth, the simulation and application protocols are tied to versioned validation assets: single-quantile TT500 experiments evaluate fit recovery and held-out forecast accuracy, while a joint multi-quantile synthetic study evaluates coherence, proper quantile scores, crossing diagnostics, and runtime under known conditional quantile paths.

The remainder of the paper is organized as follows. Section 2 introduces notation, deep echo state network (DESN) dynamics, and the exAL representation. Section 3 presents the Q–DESN model and prior specification, including the joint quantile-vector extension. Section 4 describes posterior computation, while Sections 5–6 describe posterior prediction, optional quantile synthesis, and reservoir diagnostics. Section 7 presents the simulation design, and Sections 8–9 give reproducible streamflow and electricity-price forecasting applications. Section 10 summarizes limitations and future work.

Related Work

Bayesian quantile regression provides the likelihood-based foundation for Q–DESN. Classical quantile regression targets conditional quantiles directly (Koenker and Bassett, 1978; Koenker, 2005). Bayesian implementations commonly use the asymmetric Laplace likelihood because of its connection to check-loss minimization and its mixture representation (Yu and Moyeed, 2001; Kozumi and Kobayashi, 2011). In this role, the asymmetric Laplace likelihood is a working likelihood: it defines a quantile-targeted posterior and supports conditional updates, while the coverage of model-based credible intervals may require separate assessment or adjustment under misspecification (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). The quantile-fixed generalized asymmetric Laplace distribution of Yan et al. (2025) addresses a specific limitation of the asymmetric Laplace family by allowing the error distribution to vary after the target percentile has been fixed.

Dynamic quantile models provide statistical baselines. Dynamic quantile linear models combine dynamic linear modeling with Bayesian quantile regression and provide posterior computation for time-varying quantile paths (Gonçalves et al., 2020). The exDQLM of Barata et al. (2022) adds asymmetric likelihood flexibility for dynamic quantile inference. Q–DESN changes the predictor representation: instead of specifying a low-dimensional linear dynamic state, it conditions on fixed nonlinear reservoir states and places a Bayesian quantile readout on the resulting design matrix.

The simulation comparisons with DQLM and exDQLM therefore separate the effect of the nonlinear reservoir representation from likelihood and readout-prior choices.

Reservoir computing and ESNs provide the nonlinear dynamic features used by the Q-DESN readout. ESNs were introduced by Jaeger (2001) and later reviewed within the broader reservoir-computing literature by Lukoševičius and Jaeger (2009); practical guidance is given by Lukoševičius (2012). Leaky-integrator ESNs make the leak rate an explicit time-scale parameter, while analyses of the echo state property caution that spectral-radius heuristics should be checked rather than treated as sufficient stability guarantees (Jaeger et al., 2007; Yildiz et al., 2012). Broader reservoir-computing surveys and time-series reviews discuss their role in forecasting and applied dynamic modeling (Zhang and Vargas, 2023; Yan et al., 2024; Cardoso et al., 2024; Sun et al., 2024). Deep ESNs, topology variants, and reservoir hyperparameter tuning extend the basic construction by stacking reservoirs, changing connectivity, or selecting reservoir parameters (Gallicchio et al., 2018; Kim and King, 2020; Maat et al., 2018; Bai et al., 2023; Viehweg et al., 2025). These sources motivate the fixed-feature viewpoint; Q-DESN places the Bayesian model on the quantile readout conditional on those features.

A related reservoir-computing literature studies uncertainty quantification and probabilistic forecasting. Deep ESNs with uncertainty quantification, probabilistic reservoir methods for load forecasting, activation-level uncertainty propagation, and Bayesian or state-space ESN formulations show that reservoir features can be embedded in probabilistic forecasting models (McDermott and Wikle, 2019; Guerra et al., 2023; Gandhi et al., 2018; Singh and Raman, 2025). This literature sets the boundary of the present contribution: uncertainty quantification for ESNs is already active. The narrower gap addressed here is quantile-specific Bayesian readout inference under a quantile-fixed asymmetric working likelihood with high-dimensional coefficient regularization.

Quantile-oriented ESN methods provide the closest forecasting comparison. Lv et al. (2016) use quantile-regression ESN ensembles to construct prediction intervals for blast-furnace gas generation, and Bonas et al. (2024) use penalized quantile regression to calibrate DESN forecasts for quasi-periodic climate processes. Q-DESN is complementary to these approaches. It does not treat the reservoir training or penalized-quantile calibration step as the Bayesian object; instead, conditional on the generated reservoir, it specifies an exAL readout whose posterior includes readout coefficients, likelihood scale and asymmetry, and shrinkage parameters.

Shrinkage priors enter because fixed reservoir designs can be high-dimensional and correlated. Ridge shrinkage provides the default dense Gaussian regularization. The regularized horseshoe, building on the horseshoe prior and its slab-regularized extension, is used as an adaptive global-local alternative (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, scale updates remain closed form (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). These priors regularize the readout; they should not be interpreted as guaranteeing feature-level interpretability without additional diagnostics.

Finally, a grid of quantile levels requires a distinction between post-processing and joint modeling. Separate Q-DESN fits can cross, and they do not learn dependence across quantile levels. Isotonic

regression can project the fitted grid onto the monotone cone (Barlow and Brunk, 1972), and rearrangement methods provide a general approach to noncrossing quantile and probability curves (Chernozhukov et al., 2010). These operations repair the reported quantile curve after fitting; they do not define a joint posterior over quantile-indexed coefficients. Constrained quantile-regression methods instead impose non-crossing restrictions during estimation (Wu and Liu, 2009; Bondell et al., 2010). Recent work also shows that non-crossing constraints can be viewed as fused shrinkage across quantile-specific coefficients (Szendrei et al., 2025). Bayesian joint quantile models share information across levels through process priors, score-likelihood constructions, or smoothness penalties (Yang and Tokdar, 2017; Wu and Narisetty, 2021; Wang and Cai, 2024). The QVP construction of Kohns and Szendrei (2025) is especially relevant here because it places a state-space shrinkage prior on adjacent quantile-indexed coefficient changes. The joint Q-DESN extension below adapts that idea to fixed DESN readouts and uses regularized-horseshoe shrinkage on adjacent slope innovations. The resulting prior is a soft non-crossing mechanism rather than a hard constraint; reported curves may still be projected or rearranged when exact monotonicity is required.

2 Notation and Preliminaries

The notation separates deterministic reservoir quantities from stochastic readout and likelihood quantities. Table 1 summarizes the recurring objects. Matrices and vectors are bold, as in $\mathbf{W}$ and $\mathbf{x}$; scalar observations, latent states, and parameters use standard lowercase notation unless they are named constants. The symbols IG, GIG, and $\mathcal{N}^+$ denote inverse-gamma, generalized inverse Gaussian, and positive-truncated Normal distributions. For a square matrix $\mathbf{A}$, write $\rho(\mathbf{A}) = \max_i |\lambda_i(\mathbf{A})|$, where $\{\lambda_i(\mathbf{A})\}$ are the eigenvalues. Thus $\rho(\cdot)$ denotes spectral radius, while $\rho_d \in (0, 1)$ denotes the target spectral radius for layer d . Indices use $d = 1, \dots, D$ and $t = 1, \dots, T$. When conditioning information is shown explicitly, $Q_p(y \mid \cdot)$ denotes the conditional p -quantile of the scalar response. The identity matrix, zero vector, and one vector of dimension m are denoted by $\mathbf{I}_m$, $\mathbf{0}_m$, and $\mathbf{1}_m$, respectively.

2.1 Deep Echo State Network (DESN)

The DESN is a deterministic feature map once the architecture, random seed, scaling constants, reducers, and washout convention are fixed. Conditional on these choices, posterior inference is restricted to the readout model. The main dimensions are stated before the recursion because the hierarchy contains several layer-specific matrices.

Notation and dimensions. The input vector is $\mathbf{u}_t = (y_{t-1}, \dots, y_{t-m}, \mathbf{z}_t^\top)^\top \in \mathbb{R}^{m+q}$, with m response lags and q exogenous variables. For layer d , $\mathbf{W}_d \in \mathbb{R}^{n_d \times n_d}$ is recurrent, $\mathbf{W}_1^{\text{in}} \in \mathbb{R}^{n_1 \times (m+q)}$ maps the original input to the first layer, and $\mathbf{W}_d^{\text{in}} \in \mathbb{R}^{n_d \times \tilde{n}_{d-1}}$ maps the reduced state from layer $d-1$ to layer d for $d \geq 2$. The reducer $\mathbf{Q}_{d-1} \in \mathbb{R}^{\tilde{n}_{d-1} \times n_{d-1}}$ maps $\mathbf{h}_{t,d-1} \in \mathbb{R}^{n_{d-1}}$ to $\tilde{\mathbf{h}}_{t,d-1} \in \mathbb{R}^{\tilde{n}_{d-1}}$. The readout dimension is $r = n_D + \sum_{d=1}^{D-1} \tilde{n}_d + (m+q)$, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_r)^\top \in \mathbb{R}^{r+1}$.

Symbol	Description	Dimension and support
y_t	scalar response at time t	scalar
p_0, p_k	single-fit and synthesis-grid quantile levels	$(0, 1)$
$\mathbf{z}_t$	exogenous covariates	$q \times 1$
$\mathbf{u}_t$	response lags and exogenous covariates	$(m + q) \times 1$
$\mathbf{h}_{t,d}$	reservoir state at layer d	$n_d \times 1$
$\tilde{\mathbf{h}}_{t,d}$	reduced state at layer d	$\tilde{n}_d \times 1$
α_d, ρ_d	layer leak and target spectral radius	$[0, 1]$ and $(0, 1)$
$\tilde{\mathbf{x}}_t$	feature vector without intercept	$r \times 1$
$\mathbf{x}_t$	readout design vector with intercept	$(r + 1) \times 1$
$\mathbf{X}$	fixed DESN design matrix after washout	$T \times (r + 1)$
$\boldsymbol{\beta}$	readout coefficients	$(r + 1) \times 1$
κ_β^2	default ridge slope-prior variance	positive
σ, γ	exAL scale and asymmetry parameters	$\sigma > 0, \gamma \in (L, U)$
(λ_j, τ, ζ)	local, global, and slab shrinkage scales	positive

Table 1: Key notation used throughout. Reservoir quantities are drawn or constructed once, held fixed after washout, and collected in the design matrix $\mathbf{X}$.

Random sparse weights and spectral scaling. In contrast to fully trained recurrent networks, reservoir and input weights are drawn once and kept fixed throughout inference. Let $\circ$ denote the Hadamard product. Each recurrent matrix is $\mathbf{W}_d = \Phi_d^{(W)} \circ Z_d^{(W)}$, and each input matrix is generated analogously as $\mathbf{W}_d^{\text{in}} = \Phi_d^{(\text{in})} \circ Z_d^{(\text{in})}$. Mask entries are Bernoulli with probabilities $\pi_d^{(W)}$ and $\pi_d^{(\text{in})}$; base entries are independent draws from zero-mean finite-variance distributions p_W and p_{in} , such as $\mathcal{N}(0, \sigma^2)$ or $\text{Unif}[-a, a]$. Masks and base draws are mutually independent and independent across entries and layers. The recurrent matrices are then scaled to the target spectral radii,

$$\text{Spectral-radius scaling:} \quad \bar{\mathbf{W}}_d = \frac{\rho_d}{\rho(\mathbf{W}_d)} \mathbf{W}_d, \quad \rho_d \in (0, 1).$$

If $\rho(\mathbf{W}_d) = 0$, the matrix is resampled. Input matrices $\mathbf{W}_d^{\text{in}}$ are not spectrally scaled.

DESN recursion and readout features. Figure 1 summarizes the three objects carried by the fixed feature construction: the lagged input, the layerwise reservoir update, and the readout design.

With layer-specific leak rates $\alpha_d \in [0, 1]$, elementwise nonlinearity $f(\cdot)$, and lower-layer transformation $k(\cdot)$, the fixed feature map is

$$\begin{aligned}
\text{Layer 1 innovation :} \quad & \boldsymbol{\omega}_{t,1} = f(\bar{\mathbf{W}}_1 \mathbf{h}_{t-1,1} + \mathbf{W}_1^{\text{in}} \mathbf{u}_t), \\
\text{Layer 1 update :} \quad & \mathbf{h}_{t,1} = (1 - \alpha_1) \mathbf{h}_{t-1,1} + \alpha_1 \boldsymbol{\omega}_{t,1}, \\
\text{Reduced bridge state :} \quad & \tilde{\mathbf{h}}_{t,d-1} = \mathbf{Q}_{d-1} \mathbf{h}_{t,d-1}, \quad d = 2, \dots, D, \\
\text{Layer } d \text{ innovation :} \quad & \boldsymbol{\omega}_{t,d} = f(\bar{\mathbf{W}}_d \mathbf{h}_{t-1,d} + \mathbf{W}_d^{\text{in}} \tilde{\mathbf{h}}_{t,d-1}), \quad d = 2, \dots, D, \\
\text{Layer } d \text{ update :} \quad & \mathbf{h}_{t,d} = (1 - \alpha_d) \mathbf{h}_{t-1,d} + \alpha_d \boldsymbol{\omega}_{t,d}, \quad d = 2, \dots, D, \\
\text{Readout feature stack :} \quad & \tilde{\mathbf{x}}_t = [\mathbf{h}_{t,D}^\top, k(\tilde{\mathbf{h}}_{t,1})^\top, \dots, k(\tilde{\mathbf{h}}_{t,D-1})^\top, \mathbf{u}_t^\top]^\top, \\
\text{Intercept and readout :} \quad & \mathbf{x}_t = (1, \tilde{\mathbf{x}}_t^\top)^\top, \quad \mu_t = \mathbf{x}_t^\top \boldsymbol{\beta}.
\end{aligned} \tag{2.1}$$

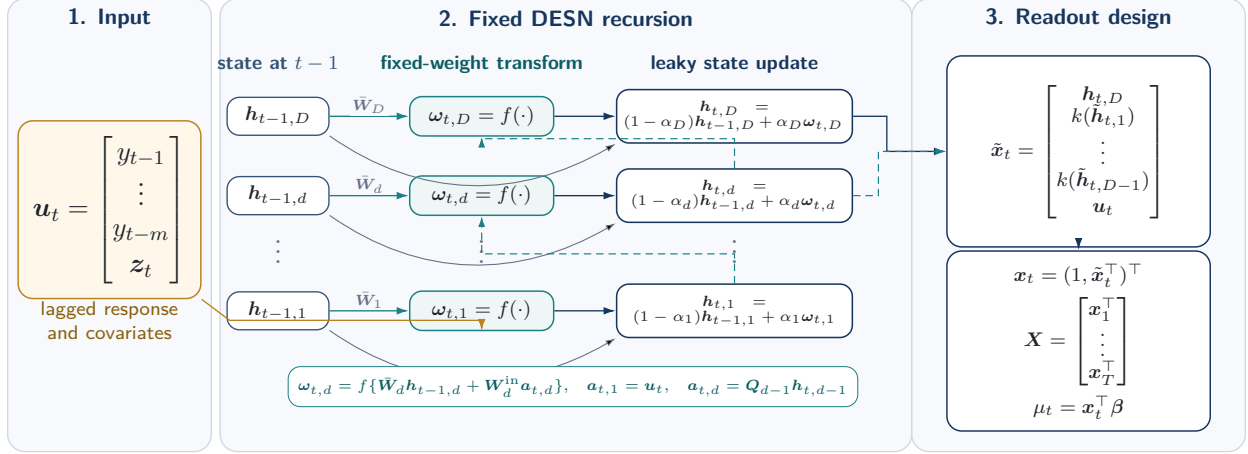


Figure 1: Fixed DESN feature construction. The input block collects response lags and exogenous covariates. The reservoir block shows that each layer update combines the previous state with a nonlinear innovation driven by fixed recurrent weights and, for upper layers, the current-time reduced state from the layer below. After washout, the top-layer state, transformed lower-layer reductions, and retained input block form the fixed readout design used by the Q-DESN likelihood.

Here $k(\cdot)$ is an elementwise activation applied to reduced lower-layer states, commonly $k = \text{id}$ or $k = f$.

The input vector $\mathbf{u}_t$ bundles response lags and exogenous covariates; the intercept is appended only in $\mathbf{x}_t$. The leaky update combines persistence $(1 - \alpha_d)\mathbf{h}_{t-1,d}$ with the nonlinear innovation $\omega_{t,d}$. The leak rate α_d and target radius ρ_d control the empirical time scale and persistence of the deterministic state recursion; they are tuning choices rather than sufficient guarantees of the echo state property. Reducers $\mathbf{Q}_{d-1}$ may be fixed random projections, optionally orthonormalized, or precomputed linear reducers such as PCA from an initial preprocessing pass.

The feature stack concatenates the top-layer state, transformed reduced lower-layer states, and $\mathbf{u}_t$, yielding $\tilde{\mathbf{x}}_t \in \mathbb{R}^r$. Appending an intercept gives $\mathbf{x}_t \in \mathbb{R}^{r+1}$, and the readout maps this feature vector to the scalar location μ_t . States are initialized at $\mathbf{h}_{0,d} = \mathbf{0}_{n_d}$, or from $\mathcal{N}(\mathbf{0}_{n_d}, \sigma_{h,d}^2 \mathbf{I}_{n_d})$. A teacher-forcing washout $t = 1, \dots, T_0$ is run before features enter the readout. Washout length and scaling choices can affect short-horizon behavior. All masks and base matrices are fixed for the entire analysis and are not resampled during posterior inference.

2.2 Extended Asymmetric Laplace (exAL)

The symbol exAL denotes the quantile-fixed generalized asymmetric Laplace (GAL) distribution of Yan et al. (2025). We use the name exAL to emphasize that it extends the asymmetric Laplace (AL) working likelihood while keeping the target quantile fixed. The standard AL likelihood is recovered as a special case, but its skewness is fixed once p_0 is chosen. The exAL asymmetry parameter allows mode, skewness, and tail behavior to vary within a quantile-fixed likelihood. At fixed level p_0 , μ is the p_0 th quantile, $\sigma > 0$ is a scale parameter, and γ is an asymmetry parameter with bounded

support (L, U) . The endpoints L and U solve $g(\gamma) = 1 - p_0$ and $g(\gamma) = p_0$, respectively, where $g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2)$, and Φ is the standard Normal CDF.

Posterior computation uses the stochastic representation of the exAL distribution. If

$$y \sim \text{exAL}_{p_0}(\mu, \sigma, \gamma),$$

then there exist independent $\epsilon \sim \mathcal{N}(0, 1)$, $z \sim \text{Exp}(1)$, and $s \sim \mathcal{N}^+(0, 1)$ such that

$$y = \mu + \lambda(\gamma)\sigma s + A(\gamma)\sigma z + \{B(\gamma)\sigma^2 z\}^{1/2}\epsilon,$$

where $A(\gamma) = \{1 - 2p_\gamma\}/\{p_\gamma(1 - p_\gamma)\}$, $B(\gamma) = 2/\{p_\gamma(1 - p_\gamma)\}$, $C(\gamma) = \{\mathbf{1}\{\gamma > 0\} - p_\gamma\}^{-1}$, $p_\gamma = \mathbf{1}\{\gamma < 0\} + \{p_0 - \mathbf{1}\{\gamma < 0\}\}/g(\gamma)$, and $\lambda(\gamma) = C(\gamma)|\gamma|$. Setting $v = \sigma z$ gives $v \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma)$ and the conditional Gaussian form used for Q-DESN posterior computation:

$$y \mid \mu, \sigma, \gamma, s, v \sim \mathcal{N}(\mu + \lambda(\gamma)\sigma s + A(\gamma)v, B(\gamma)\sigma v).$$

All quantities $g(\gamma)$, p_γ , $A(\gamma)$, $B(\gamma)$, $C(\gamma)$, and $\lambda(\gamma)$ depend on the target quantile level p_0 ; this dependence is suppressed for readability.

3 Model Specification

Fix a quantile level $p_0 \in (0, 1)$. After reservoir construction and washout, $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T)^\top$ is treated as fixed, and all posterior uncertainty is conditional on this design. The statistical model is a Bayesian quantile readout for $\mathbf{y}$ given $\mathbf{X}$. The observation model and priors define the posterior target; the Gaussian augmentation below is a computational representation of the asymmetric working likelihood. Dependence on p_0 is suppressed when no ambiguity arises.

3.1 Quantile Deep Echo State Network (Q-DESN)

The Q-DESN is obtained by inserting the fixed DESN feature map into the quantile-fixed exAL likelihood of Section 2.2. Let $\tilde{\mathbf{x}}_t$ be the feature vector in Eq. (2.1), and set $\mathbf{x}_t = (1, \tilde{\mathbf{x}}_t^\top)^\top$. Conditional on the fixed design matrix, the marginal working likelihood is

$$y_t \mid \mathbf{x}_t, \boldsymbol{\beta}, \sigma, \gamma \sim \text{exAL}_{p_0}(\mathbf{x}_t^\top \boldsymbol{\beta}, \sigma, \gamma), \quad t = 1, \dots, T.$$

Thus the dynamic readout $\mathbf{x}_t^\top \boldsymbol{\beta}$ is the model's conditional p_0 -quantile, while σ and γ control the scale and shape of the working error distribution. For posterior computation, the same exAL mixture representation is applied at each time point:

$$\begin{aligned}
\text{Quantile readout :} \quad & \mu_t = \mathbf{x}_t^\top \boldsymbol{\beta}, \quad \boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_r)^\top \in \mathbb{R}^{r+1}, \\
\text{Gaussian conditional :} \quad & y_t \mid \boldsymbol{\beta}, \sigma, \gamma, s_t, v_t, \mathbf{x}_t \sim \mathcal{N}(\mu_t + \lambda(\gamma)\sigma s_t + A(\gamma)v_t, B(\gamma)\sigma v_t), \\
\text{Positive-shift latent :} \quad & s_t \sim \mathcal{N}^+(0, 1), \\
\text{Scale-mixture latent :} \quad & v_t \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma).
\end{aligned} \tag{3.1}$$

The display separates the statistical readout from the computational augmentation. The first line defines the quantile path; the second gives the conditional Gaussian observation equation; and the last two lines introduce auxiliary variables. The constants $A(\gamma)$, $B(\gamma)$, and $\lambda(\gamma)$ are inherited from Section 2.2. The DESN feature map is fixed before posterior inference, while $\boldsymbol{\beta}$, $\sigma > 0$, and $\gamma \in (L, U)$ are posterior unknowns. Integrating out (s_t, v_t) recovers the marginal exAL likelihood, so these variables do not alter the quantile target; they expose conditionally Gaussian updates. Complete-data joint distributions for the AL and exAL likelihoods with ridge and RHS priors are given in Section S4 of the supplement.

3.2 Joint Quantile-Vector Q-DESN

The single-level model can be fit independently over a grid of quantile levels, but independent fitting does not couple the readout coefficients across levels. For applications in which the coefficient path itself is part of the inferential target, let $0 < p_1 < \dots < p_Q < 1$ be a fixed quantile grid and write $\mathbf{Z}$ for the $T \times r$ DESN design matrix without the intercept column. The joint quantile-vector readout separates intercepts from high-dimensional slopes:

$$Q_{p_q}(y_t \mid \mathcal{F}_t) = \alpha_q + \tilde{\mathbf{x}}_t^\top \boldsymbol{\theta}_q, \quad q = 1, \dots, Q, \quad t = 1, \dots, T, \tag{3.2}$$

where $\alpha_q \in \mathbb{R}$ is the quantile-specific intercept and $\boldsymbol{\theta}_q \in \mathbb{R}^r$ is the DESN slope readout at level p_q . The intercepts are not included in the high-dimensional RHS slope path by default; they are either weakly regularized independently or represented through ordered gaps when a stronger monotonicity convention is desired.

For each q , the working likelihood uses the quantile-fixed exAL family at the corresponding target level,

$$y_t \mid \alpha_q, \boldsymbol{\theta}_q, \sigma_q, \gamma_q \sim \text{exAL}_{p_q}(\alpha_q + \tilde{\mathbf{x}}_t^\top \boldsymbol{\theta}_q, \sigma_q, \gamma_q). \tag{3.3}$$

The Q likelihood contributions form a composite working likelihood over the same response series. A learning-rate parameter $\kappa > 0$ may be used to temper this composite contribution, with $\kappa = 1$ as the default and $\kappa = 1/Q$ as a conservative sensitivity setting under the general Bayes view of loss-based updating (Bissiri et al., 2016).

The joint prior follows the QVP idea that adjacent quantile-indexed coefficient changes should be small unless the data support tail-specific effects (Kohns and Szendrei, 2025). Let $\boldsymbol{\theta}_0 \in \mathbb{R}^r$ be a

baseline readout and define adjacent innovations

$$\Delta_{1,j} = \theta_{1,j} - \theta_{0,j}, \quad \Delta_{q,j} = \theta_{q,j} - \theta_{q-1,j}, \quad q = 2, \dots, Q.$$

The baseline slope receives an ordinary RHS prior, while the adjacent innovations receive RHS shrinkage with quantile-gap-specific global scales:

$$\theta_{0,j} \mid \lambda_j^0, \tau^0, \zeta_0 \sim \mathcal{N}(0, V_j^0), \quad V_j^0 = \frac{\zeta_0^2 (\tau^0)^2 (\lambda_j^0)^2}{\zeta_0^2 + (\tau^0)^2 (\lambda_j^0)^2}, \quad (3.4)$$

$$\Delta_{q,j} \mid \lambda_{q,j}^\Delta, \tau_q^\Delta, \zeta_\Delta \sim \mathcal{N}(0, V_{q,j}^\Delta), \quad V_{q,j}^\Delta = \frac{\zeta_\Delta^2 (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}{\zeta_\Delta^2 + (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}. \quad (3.5)$$

This construction separates globally useful reservoir features θ_0 from features whose effects vary across adjacent quantile levels. It is related to interquantile shrinkage and composite-quantile regularization in frequentist quantile regression (Jiang et al., 2014; Zou and Yuan, 2008), but the shrinkage is placed on the Bayesian readout path conditional on fixed reservoir features.

Let $\theta = (\theta_1^\top, \dots, \theta_Q^\top)^\top$, $\mathbf{Z}_{\text{st}} = \mathbf{I}_Q \otimes \mathbf{Z}$, and $\alpha_{\text{st}} = (\alpha_1 \mathbf{1}_T^\top, \dots, \alpha_Q \mathbf{1}_T^\top)^\top$. Under the exAL augmentation, define

$$d_{q,t} = \lambda(\gamma_q) \sigma_q s_{q,t} + A(\gamma_q) v_{q,t}, \quad \omega_{q,t} = B(\gamma_q) \sigma_q v_{q,t},$$

and stack these quantities into $\mathbf{d}$ and diagonal $\mathbf{\Omega} = \text{diag}(\omega_{q,t})$. The conditional Gaussian likelihood is

$$\mathbf{y}_{\text{st}} \mid \theta, \alpha, \mathbf{d}, \mathbf{\Omega} \sim \mathcal{N}(\alpha_{\text{st}} + \mathbf{Z}_{\text{st}} \theta + \mathbf{d}, \mathbf{\Omega}), \quad \mathbf{y}_{\text{st}} = \mathbf{1}_Q \otimes \mathbf{y}.$$

Let $\mathbf{H}$ be the block first-difference matrix such that

$$\mathbf{H}\theta = (\theta_1^\top, (\theta_2 - \theta_1)^\top, \dots, (\theta_Q - \theta_{Q-1})^\top)^\top.$$

With $\tilde{\theta}_0 = (\theta_0^\top, \mathbf{0}_r^\top, \dots, \mathbf{0}_r^\top)^\top$ and diagonal $\mathbf{D}_\Delta$ containing the RHS innovation variances $V_{q,j}^\Delta$, the conditional prior is proportional to

$$\exp \left[-\frac{1}{2} (\mathbf{H}\theta - \tilde{\theta}_0)^\top \mathbf{D}_\Delta^{-1} (\mathbf{H}\theta - \tilde{\theta}_0) \right].$$

Consequently, the Gaussian full conditional for the stacked slope path has precision

$$\mathbf{K}_\theta = \kappa \mathbf{Z}_{\text{st}}^\top \mathbf{\Omega}^{-1} \mathbf{Z}_{\text{st}} + \mathbf{H}^\top \mathbf{D}_\Delta^{-1} \mathbf{H}, \quad (3.6)$$

and mean

$$\mathbf{m}_\theta = \mathbf{K}_\theta^{-1} \left[\kappa \mathbf{Z}_{\text{st}}^\top \mathbf{\Omega}^{-1} (\mathbf{y}_{\text{st}} - \alpha_{\text{st}} - \mathbf{d}) + \mathbf{H}^\top \mathbf{D}_\Delta^{-1} \tilde{\theta}_0 \right]. \quad (3.7)$$

The sparse, block-banded prior contribution in (3.6) is the central computational object for the joint extension.

The non-crossing interpretation is local in adjacent quantile gaps. For neighboring levels,

$$Q_{p_q}(y_t \mid \mathcal{F}_t) - Q_{p_{q-1}}(y_t \mid \mathcal{F}_t) = (\alpha_q - \alpha_{q-1}) + \tilde{\mathbf{x}}_t^\top (\boldsymbol{\theta}_q - \boldsymbol{\theta}_{q-1}).$$

If each component of the scaled feature vector lies in $[-1, 1]$, then $\alpha_q - \alpha_{q-1} \geq \|\boldsymbol{\theta}_q - \boldsymbol{\theta}_{q-1}\|_1$ is a sufficient condition for non-crossing over that bounded feature domain. The adjacent-difference RHS prior does not impose this inequality. Instead, it shrinks $\boldsymbol{\theta}_q - \boldsymbol{\theta}_{q-1}$ toward zero, reducing the size of the feature-dependent term and concentrating the posterior near coherent quantile paths. Exact monotonicity still requires an explicit ordered-intercept parameterization, posterior rejection rule, isotonic projection, or monotone rearrangement.

3.3 Prior Specification

The intercept is modeled separately from the slope coefficients: $\beta_0 \sim \mathcal{N}(b_0, V_0)$, with $b_0 \in \mathbb{R}$ and $V_0 > 0$. This separation avoids forcing the baseline quantile level toward zero when the readout features are not centered around the target quantile.

For the non-intercept coefficients $\boldsymbol{\beta}_{1:r} = (\beta_1, \dots, \beta_r)^\top$, the default ridge prior is

$$\boldsymbol{\beta}_{1:r} \mid \kappa_\beta^2 \sim \mathcal{N}(\mathbf{0}_r, \kappa_\beta^2 \mathbf{I}_r), \quad (3.8)$$

where $\kappa_\beta^2 > 0$ is fixed or assigned a weakly informative scale prior. This Gaussian prior gives dense baseline regularization.

The regularized horseshoe (RHS) prior is the adaptive shrinkage alternative. It starts from the ordinary horseshoe (Carvalho et al., 2010), adds slab regularization (Piironen and Vehtari, 2017), and uses the product representation of Nishimura and Suchard (2023) to retain closed-form scale updates. For non-intercept coefficients $j = 1, \dots, r$, the ordinary horseshoe is

$$\beta_j \mid \tau, \lambda_j \sim \mathcal{N}(0, \tau^2 \lambda_j^2), \quad \lambda_j \sim C^+(0, 1), \quad \tau \sim C^+(0, \tau_0), \quad (3.9)$$

where τ is the global scale and λ_j is coefficient-specific. This prior concentrates mass near zero while allowing some coefficients to escape strong shrinkage, but it does not separately control the largest prior variances. The regularized version introduces a slab scale $\zeta > 0$ and sets

$$V_j(\lambda_j, \tau, \zeta) = \left(\zeta^{-2} + \tau^{-2} \lambda_j^{-2} \right)^{-1} = \frac{\zeta^2 \tau^2 \lambda_j^2}{\zeta^2 + \tau^2 \lambda_j^2}, \quad j = 1, \dots, r, \quad (3.10)$$

with conditional prior

$$\beta_j \mid \lambda_j, \tau, \zeta \sim \mathcal{N}(0, V_j), \quad j = 1, \dots, r. \quad (3.11)$$

This is the RHS coefficient prior used in the adaptive-shrinkage Q-DESN fits. Equation (3.10) gives the interpretation: if $\tau^2 \lambda_j^2 \ll \zeta^2$, then $V_j \approx \tau^2 \lambda_j^2$, recovering ordinary horseshoe behavior; if $\tau^2 \lambda_j^2 \gg \zeta^2$, then $V_j \approx \zeta^2$, so the slab caps the effective prior variance.

The direct Piironen–Vehtari joint formulation gives the desired regularized horseshoe coefficient law. For Gibbs and CAVI updates, however, its global-local block includes an extra $\{1 + \tau^2 \lambda_j^2 / \zeta^2\}^{1/2}$ factor, so the ordinary horseshoe updates for (τ, λ_j) are not preserved. The Nishimura–Suchard joint construction preserves the same conditional coefficient law while restoring closed-form scale updates:

$$p(\beta_j, \lambda_j \mid \tau, \zeta) \propto \exp\left(-\frac{\beta_j^2}{2\tau^2 \lambda_j^2}\right) \exp\left(-\frac{\beta_j^2}{2\zeta^2}\right) \lambda_j^{-1} (1 + \lambda_j^2)^{-1}, \quad (3.12)$$

where proportionality is with respect to (β_j, λ_j) . The (τ, ζ) -dependent Gaussian normalizing factors are retained in the full-conditionals derivations so that the inverse-gamma shape parameters are recovered correctly. Thus the RHS option adds adaptive global-local shrinkage without losing inverse-gamma updates for the local and global scales.

Closed-form scale updates use the inverse-gamma representation of half-Cauchy priors (Makalic and Schmidt, 2016):

$$\lambda_j^2 \mid \nu_j \sim \text{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \quad \nu_j \sim \text{IG}\left(\frac{1}{2}, 1\right), \quad j = 1, \dots, r, \quad (3.13)$$

$$\tau^2 \mid \xi \sim \text{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \quad \xi \sim \text{IG}\left(\frac{1}{2}, \frac{1}{\tau_0^2}\right). \quad (3.14)$$

The slab scale is either fixed or assigned

$$\zeta^2 \sim \text{IG}(a_\zeta, b_\zeta). \quad (3.15)$$

Under (3.12), this prior remains compatible with closed-form updating (Nishimura and Suchard, 2023).

The likelihood parameters are assigned $\sigma \sim \text{IG}(a_\sigma, b_\sigma)$ and $\gamma \sim \pi_\gamma(\gamma) \mathbf{1}\{L < \gamma < U\}$, with (L, U) determined by the quantile-fixing construction as described in Section 2.2. For RHS fits, collect $\Theta = (\beta, \sigma, \gamma, \boldsymbol{\lambda}^2, \tau^2, \boldsymbol{\nu}, \xi, \zeta^2)$, where $\boldsymbol{\lambda}^2 = (\lambda_1^2, \dots, \lambda_r^2)^\top$ and $\boldsymbol{\nu} = (\nu_1, \dots, \nu_r)^\top$. The entry ζ^2 is omitted when the slab scale is fixed. Ridge fits replace the global-local shrinkage block by κ_β^2 when that scale is assigned a prior.

4 Posterior Inference

Let Θ collect the readout coefficients, exAL parameters, and prior scale parameters, and let $\mathbf{a} = (\mathbf{v}, \mathbf{s})$ collect the exAL auxiliary variables. Conditional on the fixed DESN design $\mathbf{X}$, the augmented posterior target is

$$p(\Theta, \mathbf{a} \mid \mathbf{y}, \mathbf{X}) \propto \left\{ \prod_{t=1}^T p(y_t \mid \beta, \sigma, \gamma, v_t, s_t, \mathbf{x}_t) p(v_t \mid \sigma) p(s_t) \right\} p(\Theta),$$

with the s_t and γ terms omitted under the AL special case. The reduced Q-DESN with the AL likelihood is conditionally conjugate under this augmentation: the readout, likelihood scale, and shrinkage-scale blocks have standard closed-form Gibbs updates. The exAL extension preserves the Gaussian readout structure, but the additional asymmetry parameter γ enters $A(\gamma)$, $B(\gamma)$, and $\lambda(\gamma)$, so the scale-asymmetry block (σ, γ) is no longer conjugate. We therefore use two computation routes for this same fixed-design target: an MCMC sampler with a separate update for the exAL block and a mean-field variational Bayes approximation.

4.1 Markov Chain Monte Carlo

The augmented likelihood and shrinkage hierarchy give a partially Gibbs sampler. Blocks with named full conditionals are updated directly; the complete-data targets, full conditionals, CAVI factors, and ELBO terms for AL and exAL likelihoods under ridge and RHS priors are given in Sections S4–S7 of the supplement. Under the Nishimura–Suchard hierarchy, the global-local scales have inverse-gamma updates. The exAL block (σ, γ) remains non-conjugate and is updated with Metropolis–Hastings or slice moves on transformed coordinates. Algorithm 1 summarizes the exAL sampler with regularized horseshoe shrinkage. Ridge and AL variants omit the corresponding shrinkage, s_t , and γ blocks. Diagnostics use trace plots, effective sample size summaries, and monitoring of the (σ, γ) update, including acceptance rates when Metropolis–Hastings is used.

Algorithm 1 MCMC sampler for Q-DESN with exAL and regularized horseshoe shrinkage. AL and ridge variants omit the corresponding s_t , γ , and shrinkage blocks.

Require: Data $\{y_t, \mathbf{x}_t\}_{t=1}^T$, priors, MCMC iterations N

- 1: Initialize (β, σ, γ) , shrinkage scales, and latents $(v_{1:T}, s_{1:T})$
 - 2: **for** $i = 1$ to N **do**
 - 3: Sample $v_{1:T}$ from independent generalized inverse Gaussian (GIG) full conditionals
 - 4: Sample $s_{1:T}$ from independent positive-truncated Normal full conditionals
 - 5: Sample β from its multivariate Normal conditional
 - 6: Sample $\{\lambda_j^2, \nu_j\}_{j=1}^r$, τ^2 , and ξ from inverse-gamma (IG) conditionals
 - 7: If ζ^2 is random, sample it from its IG conditional; otherwise keep it fixed
 - 8: Update (σ, γ) by Metropolis–Hastings (MH) or slice sampling
 - 9: Store post-burn-in draws, with thinning only if used
 - 10: **end for**
-

4.2 Variational Bayes

For lower-cost approximate inference, we use a mean-field variational Bayes approximation for the augmented posterior above. It is an approximation to the same fixed-design model, not a different model. The Gaussian, GIG, positive-truncated Normal, and inverse-gamma factors have closed-form coordinate updates. The exAL scale-asymmetry block is non-conjugate. Building on the variational strategy of Barata et al. (2022) and the non-conjugate variational ideas of Wang and Blei

(2013), the VB–LD implementation uses a Laplace approximation for this block and Delta-method approximations for expectations of smooth functions of (σ, γ) . This replaces an importance-sampling step with deterministic approximations inside the coordinate updates.

For Q–DESN under the regularized horseshoe formulation above, with $\mathbf{a} = (\mathbf{v}, \mathbf{s})$, the variational factorization is

$$q(\Theta, \mathbf{a}) = q_{\beta}(\boldsymbol{\beta}) q_{\sigma, \gamma}(\sigma, \gamma) q_{\tau^2}(\tau^2) q_{\xi}(\xi) \prod_{j=1}^r q_{\lambda_j^2}(\lambda_j^2) q_{\nu_j}(\nu_j) q_{\zeta^2}(\zeta^2) \prod_{t=1}^T q_{v_t}(v_t) q_{s_t}(s_t),$$

where q_{ζ^2} is omitted if ζ is fixed. The factors for $\boldsymbol{\beta}$, $\{v_t\}$, $\{s_t\}$, $\{\lambda_j^2\}$, $\{\nu_j\}$, τ^2 , and ξ remain in Gaussian, GIG, positive-truncated Normal, and inverse-gamma families. Only $q_{\sigma, \gamma}$ is approximated with Laplace–Delta on unconstrained coordinates. Posterior means and predictive summaries from VB–LD are therefore variational approximations, with additional approximation error from this Laplace–Delta block.

The Laplace–Delta step uses unconstrained coordinates $\boldsymbol{\eta} = (\log \sigma, \log\{(\gamma - L)/(U - \gamma)\})^\top$, on which the expected complete-data log posterior, including the Jacobian, is approximated by a Normal density at its mode. If $\hat{\boldsymbol{\eta}}$ is the mode and Σ is the negative inverse Hessian at that mode, then smooth expectations needed by the closed-form updates are approximated by

$$\mathbb{E}\{g(\boldsymbol{\eta})\} \approx g(\hat{\boldsymbol{\eta}}) + \frac{1}{2} \text{Tr}\{\nabla^2 g(\hat{\boldsymbol{\eta}}) \Sigma\}.$$

Thus, only expectations involving the exAL scale and asymmetry use Laplace–Delta; the remaining factors are updated in closed form. The updates are iterated until the evidence lower bound (ELBO), or its Laplace–Delta approximation, stabilizes.

Algorithm 2 outlines the coordinate ascent updates for the exAL and regularized horseshoe case. Ridge and AL variants omit the corresponding blocks. The output is a set of approximate posterior moments and predictive summaries at the target quantile levels.

Algorithm 2 Variational Bayes updates for Q-DESN with exAL and regularized horseshoe shrinkage. AL and ridge variants omit the corresponding latent, asymmetry, and shrinkage factors.

Require: Data $\{y_t, \mathbf{x}_t\}_{t=1}^T$, priors, convergence controls

- 1: Initialize factors for $q(\boldsymbol{\beta})$, $q(v_{1:T})$, $q(s_{1:T})$, shrinkage scales, and $q(\sigma, \gamma)$
 - 2: **repeat**
 - 3: Update $q(v_{1:T})$ as independent generalized inverse Gaussian (GIG) factors
 - 4: Update $q(s_{1:T})$ as independent positive-truncated Normal factors
 - 5: Update $q(\boldsymbol{\beta})$ as a multivariate Normal factor
 - 6: Update $\{q(\lambda_j^2), q(\nu_j)\}_{j=1}^r$, $q(\tau^2)$, and $q(\xi)$ as inverse-gamma (IG) factors
 - 7: If ζ^2 is random, update $q(\zeta^2)$ as an IG factor; otherwise keep it fixed
 - 8: Update $q(\sigma, \gamma)$ by a Laplace approximation on unconstrained coordinates
 - 9: Update Delta-method expectations for smooth functions of (σ, γ)
 - 10: Evaluate the ELBO or Laplace–Delta ELBO approximation and check convergence
 - 11: **until** convergence
-

4.3 Joint Quantile-Vector Inference

The joint quantile-vector extension targets the augmented posterior defined by the composite exAL likelihood in (3.3), the intercept prior, the baseline RHS prior (3.4), and the innovation RHS prior (3.5). Conditional on the exAL latent variables and shrinkage scales, the only structural change from the single-level sampler is that the readout update becomes the stacked Gaussian update in (3.6)–(3.7). The latent variables $\{v_{q,t}, s_{q,t}\}$ and likelihood parameters (σ_q, γ_q) are updated by quantile level, while the adjacent-difference RHS prior couples the slope vectors $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_Q$.

The MCMC sampler alternates between five groups: exAL latent variables for each q , the stacked slope path $\boldsymbol{\theta}$, the baseline readout $\boldsymbol{\theta}_0$, intercepts $\boldsymbol{\alpha}$, and the baseline and innovation RHS scale blocks. The intercept block may be updated as an unconstrained Gaussian block or through ordered gaps if exact intercept monotonicity is part of the reported specification. The scale-asymmetry block (σ_q, γ_q) remains non-conjugate and is updated separately for each q on transformed coordinates.

The VB–LD approximation uses the same factorization logic as the single-level model, with multivariate Gaussian factors for $\boldsymbol{\theta}$, $\boldsymbol{\theta}_0$, and $\boldsymbol{\alpha}$, inverse-gamma factors for the RHS scales, GIG factors for the $v_{q,t}$, positive-truncated Normal factors for the $s_{q,t}$, and independent Laplace–Delta factors for the quantile-specific (σ_q, γ_q) blocks. The resulting ELBO is exact for the conditionally conjugate Gaussian, GIG, truncated-normal, and inverse-gamma factors, and approximate for smooth functions of the Laplace–Delta blocks. Section S8 of the supplement gives the full posterior factorization, conditional updates, variational factorization, ELBO terms, and crossing diagnostics.

5 Posterior Prediction and Quantile Synthesis

5.1 Multi-Step Posterior Prediction

Forecasting is conditional on the fitted readout, the fixed reservoir construction, and the working likelihood. At a forecast origin T , the historical reservoir state is obtained by teacher forcing: the recursion in Eq. (2.1) is run through the observed record with observed responses in the lag vector $\mathbf{u}_t$. After T , future responses are unobserved, so multi-step forecasting is draw-specific. MCMC propagates posterior draws of $(\boldsymbol{\beta}, \sigma, \gamma)$, whereas VB-LD propagates draws or moments from the variational approximation.

For a draw indexed by m , initialize the reservoir at its stored state at time T . At horizon $h = 1, \dots, H$, form $\mathbf{u}_{T+h}^{(m)}$ from origin-available covariates and response lags. Lags whose times are at most T use observed responses; lags that fall inside the forecast window use previously simulated values from the same draw. The reservoir is then updated with Eq. (2.1), giving a draw-specific design vector $\mathbf{x}_{T+h}^{(m)}$. The working-likelihood forecast draw is

$$y_{T+h}^{(m)} \sim \text{exAL}_{p_0} \left(\mathbf{x}_{T+h}^{(m)\top} \boldsymbol{\beta}^{(m)}, \sigma^{(m)}, \gamma^{(m)} \right).$$

The simulated value $y_{T+h}^{(m)}$ is then inserted into future lag vectors for that same path. Repeating this recursion over posterior or variational draws yields working-likelihood forecast draws, model-based forecast bands, predictive quantile summaries, and empirical coverage diagnostics when validation outcomes are available. Exogenous covariates must be known at the forecast origin, supplied as scenarios, or generated by an external covariate model. Additional details are given in Section S9 of the supplement.

5.2 Post Hoc Quantile Synthesis

The Q-DESN likelihood is specified at one quantile level at a time. When models are fit independently on a grid $p_1 < \dots < p_K$, their fitted values are target-level quantile summaries, not draws from a joint posterior over a quantile function. Post hoc synthesis is useful only when a monotone predictive quantile curve is needed for interval construction, display, or distributional forecast evaluation. The same construction applies to any quantile forecasting method that returns ordered-level summaries; here it remains an optional post-processing step rather than a component of the fitted Q-DESN likelihood.

For each level p_k , let $\hat{q}_{k,t}$ denote a posterior or variational summary of the target quantile at time t , such as the posterior mean or median of $\mu_{k,t} = \mathbf{x}_t^\top \boldsymbol{\beta}_k$, or the corresponding forecast-horizon summary from the recursion in Section 5.1. The synthesis uses the target grid $\{(p_k, \hat{q}_{k,t})\}_{k=1}^K$, not off-target quantiles implied by the exAL_{p_k} working likelihood. Since independent fits can cross, first

Workflow	What is coupled	Interpretation
Independent fits with isotonic or rearrangement synthesis	Only the reported quantile grid is projected or rearranged after separate single-level fits.	Useful for interval construction, display, and distributional scoring when the fitted target levels need a monotone reported curve; not a joint posterior over quantile-indexed readouts.
Joint quantile-vector readout with adjacent RHS shrinkage	Adjacent slope innovations $\theta_q - \theta_{q-1}$ are shrunk during posterior inference, with intercepts handled separately.	Useful when coefficient-path coherence across quantile levels is part of the modeling target; gives soft non-crossing behavior and may still be followed by projection or rearrangement for exact monotone reporting.

Table 2: Two multi-quantile Q-DESN reporting workflows.

project the grid onto the monotone cone,

$$\mathbf{q}_t^* = \arg \min_{q_1 \leq \dots \leq q_K} \sum_{k=1}^K (\hat{q}_{k,t} - q_k)^2, \quad (5.1)$$

using isotonic regression (Barlow and Brunk, 1972). With $\mathbf{q}_t^* = (\hat{q}_{1,t}^*, \dots, \hat{q}_{K,t}^*)$, define the estimated predictive quantile function by piecewise linear interpolation:

$$Q_t^{\text{synth}}(u) = \begin{cases} \hat{q}_{1,t}^*, & u \leq p_1, \\ (1-w)\hat{q}_{k,t}^* + w\hat{q}_{k+1,t}^*, & p_k \leq u < p_{k+1}, \\ \hat{q}_{K,t}^*, & u \geq p_K, \end{cases} \quad (5.2)$$

where $k = \max\{j : p_j \leq u\}$ and $w = (u - p_k)/(p_{k+1} - p_k)$ in the middle line. Endpoint behavior is intentionally conservative: outside the fitted quantile range, the grid does not identify tail shape. If draw-level curves or smoother interpolants are used and residual non-monotonicity appears on a dense evaluation grid, a final monotone rearrangement can be applied before interpolation (Chernozhukov et al., 2010). The simulation study below evaluates fitted quantile levels directly; synthesis is invoked only when a monotone multi-quantile forecast summary is reported.

5.3 Independent Synthesis and Joint Quantile-Vector Reporting

The two multi-quantile workflows answer different questions and should not be interpreted interchangeably. Table 2 summarizes the distinction. Independent fits preserve the single-target posterior interpretation at each p_k and use synthesis only to report a monotone curve. The joint quantile-vector readout instead places a prior on the coefficient path during inference, so information is shared across adjacent quantile levels before prediction summaries are formed.

6 Model Diagnostics and Reservoir Specification

Reservoir tuning, numerical diagnosis, and forecast evaluation have distinct roles. Reservoir tuning chooses a reproducible feature-generation rule before the final analysis; numerical diagnostics screen candidate feature maps for alignment, stability, redundancy, and conditioning failures; forecast evaluation is defined by the simulation or application protocol. This section therefore gives a selection and diagnostic workflow. The simulation and streamflow sections specify the concrete fit-recovery and, where applicable, forecast-scoring criteria used in their empirical designs.

Let

$$\mathcal{R} = \{D, \{n_d\}, \{\tilde{n}_d\}, m, T_0, \{\alpha_d\}, \{\rho_d\}, \{\pi_d^{(W)}\}, \{\pi_d^{(\text{in})}\}, f, k, \text{seed}\}$$

denote the depth, layer sizes, reduction sizes, lag memory, washout, leak rates, spectral radii, sparsity probabilities, activations, and random seed used to construct Eq. (2.1). Once $\mathcal{R}$ and the preprocessing convention are fixed, posterior inference is conditional on the resulting design matrix $\mathbf{X}$. Specification selection chooses among candidate feature maps before the reported analysis; it does not average over reservoir architectures in the posterior target.

The ESN literature treats these choices as task-dependent design parameters. Reservoir size controls feature richness, leak rates and spectral radii jointly affect memory and stability, and input scaling determines how strongly the reservoir is driven into nonlinear regimes (Jaeger et al., 2007; Lukoševičius, 2012). For deep reservoirs, stacking layers can create hierarchical temporal representations, but additional depth must be justified by validation performance and stability diagnostics rather than by architecture alone (Gallicchio et al., 2018). The condition $\rho_d < 1$ is a useful conservative design choice, but it is not by itself a complete empirical check of the echo state property for a driven nonlinear reservoir (Yildiz et al., 2012). Several ESN modification papers identify failure modes that are visible before forecast scoring: output weights can reveal task-relevant reservoir units, state feedback changes the effective recurrent dynamics through an external feedback channel, redundant state trajectories can damage readout conditioning, and random reservoir realizations can induce substantial seed-to-seed variability (Fan et al., 2017; Ehlers et al., 2025; Saadat et al., 2025; Wu et al., 2018). In Q-DESN these results enter as diagnostics, sensitivity analyses, and possible ablations. They do not change the inferential target: after the feature map passes the validation protocol, the reservoir is fixed for posterior inference.

The protocol begins by fixing the empirical design before candidate reservoirs are compared: training and validation windows, held-out test windows, forecast origins, forecast horizons, quantile levels, input transformations, and scoring rules. Each candidate then passes a design check that verifies input alignment, feature dimensions, washout handling, finite state values, empirical spectral radii, state saturation, and readout conditioning. State-matrix diagnostics include near-zero-variance columns, pairwise state correlations, the singular-value or covariance spectrum, effective rank, and a condition number or eigenvalue-spread summary. These summaries distinguish a large reservoir that adds nonredundant dynamics from one that mainly adds collinear features. Candidates that fail the check are excluded before score ranking or sent to a remedial path, such as stronger shrinkage,

Component	Role in the fixed feature map	Selection rule	Main diagnostic risk
D	Number of reservoir layers and hierarchy of time scales	Search over a small set and retain additional depth only when validation scores improve after seed checks.	Unnecessary depth can add unstable or redundant features.
$\{n_d\}, \{\tilde{n}_d\}$	Reservoir capacity and lower-layer feature dimension	Treat as the main complexity control, with reduction sizes selected as part of the same feature map.	Large designs can overfit or produce poorly conditioned readouts; redundant states can reduce the effective rank of $\mathbf{X}$.
m	Explicit lag memory entering the reservoir input	Choose a domain-informed grid and require all lags to be available at the forecast origin.	Excessive lags increase dimension and can leak future information if alignment is not audited.
$\{\alpha_d\}, \{\rho_d\}$	Reservoir time scale, persistence, and stability	Co-tune because both affect memory. Near-one radii require empirical stability checks.	Saturation, state collapse, or sensitivity to initial conditions.
$\{\pi_d^{(W)}\}, \{\pi_d^{(\text{in})}\}$ and input scaling	Internal connectivity and input excitation	Fix from a small set of literature-guided defaults; broaden the search when diagnostics show under-excitation or saturation.	Too sparse can reduce richness; too dense or strongly driven can saturate nonlinear states.
f, k, T_0	Nonlinearity, bridge transformation, and initial-condition washout	Fix before reservoir selection, with T_0 chosen relative to the memory scale.	Short washout can leave initial-condition artifacts.
Seed	Realization of the random feature map	Treat as a robustness variable, not as a hidden tuning parameter. Refit top candidates over multiple seeds.	A single selected seed can overstate validation performance.

Table 3: Reservoir specification components for the fixed Q-DESN feature map. The selected object is the design used by the Bayesian readout, so selection is based on validation evidence and design diagnostics rather than posterior uncertainty over reservoir weights.

dimension reduction, or state pruning.

Candidate ranking should combine design diagnostics with validation scores, but the relevant scores depend on the empirical setting. In simulations with a known oracle quantile path, fit-recovery metrics such as quantile-path RMSE, mean absolute error, bias, and correlation are available. In both simulations and applications, forecast scoring must use common forecast origins and horizons for all candidates. For a single fitted quantile level, pinball loss is the proper score aligned with the target functional. When several levels are fit and an estimated predictive quantile function is reported, interval score, empirical coverage, interval length, CRPS from the synthesized grid, and PIT summaries may also be used, provided the fitted grid is dense enough for the summary being claimed. These scores evaluate the forecast distribution induced by the fitted working-likelihood models; they do not by themselves establish calibrated posterior credible bands under likelihood misspecification.

The coarse search focuses on the primary design parameters in Table 3: D , $\{n_d\}$, $\{\tilde{n}_d\}$, m , $\{\alpha_d\}$, and $\{\rho_d\}$. Secondary choices such as recurrent sparsity, input sparsity, input scaling, and activations are broadened only when diagnostics indicate a specific failure mode. Top candidates are refit over multiple reservoir seeds and, when computationally feasible, compared under both VB-LD and MCMC on a smaller diagnostic subset. The final reservoir specification is frozen before evaluation on held-out test origins. If reservoir ensembles are used, they aggregate only reservoirs that passed the same design checks and are reported as an ensemble sensitivity analysis rather than as posterior uncertainty conditional on a single fixed feature map.

Reproducible reporting includes the candidate grid, fixed defaults, state-quality summaries, rejected candidates and rejection reasons, validation origins, forecast horizons, scoring metrics, reservoir seeds, package and code versions, and the final selected $\mathcal{R}$. Claims about a selected architecture are conditional on this validation design. A single seed, a single forecast origin, or one favorable metric is not evidence of global optimality for the reservoir architecture. Reservoir-adaptation mechanisms such as state feedback or principal-neuron reinforcement are useful future extensions, but they change the feature-generation rule and require their own validation protocol before being combined with the Q-DESN readout.

7 Simulation Validation Study

The simulation section has two complementary parts. The first is a fixed-size single-quantile benchmark, referred to as TT500, that follows the family-based comparison strategy of Yan et al. (2025). The target quantile level is fixed in advance, the data-generating mechanism is known, and competing models are compared under matched error families, fit-window sizes, and quantile levels. The second is a joint multi-quantile validation study that uses known conditional quantiles to compare joint and independent QDESN and exQDESN readouts over a fixed grid of quantile levels.

In contrast to the linear-regression setting of Yan et al. (2025), the oracle quantile paths are dynamic, and Q-DESN represents those paths through high-dimensional deterministic reservoir

features. The design therefore separates two questions: how accurately a fitted model recovers the known conditional quantile path, and how well it forecasts held-out dynamic blocks. We write p for the single target level in the TT500 study and τ for levels in the multi-quantile grid.

The reported tables are generated from verified source files. The table builders check source hashes, rolling-origin metadata, fit and forecast windows, and table hashes before writing manuscript assets. Run-level provenance remains in the generated manifests. The larger TT5000 MCMC study remains outside the present simulation tables and requires a separate pinned validation interface. The joint multi-quantile validation below has its own registry, table assets, and crossing-diagnostic contract.

7.1 Single-Quantile TT500 Data-Generating Processes

For each error family f and target quantile level $p \in \{0.05, 0.25, 0.50\}$, the synthetic series is generated as

$$y_{t,p,f} = \mu_{t,f} + \varepsilon_{t,p,f}, \quad \varepsilon_{t,p,f} = e_{t,f} - H_f^{-1}(p),$$

where H_f is the distribution function of the raw error $e_{t,f}$. The shift by $H_f^{-1}(p)$ makes the p -quantile of $\varepsilon_{t,p,f}$ equal to zero. Hence

$$Q_p(y_{t,p,f} \mid \boldsymbol{\theta}_{t,f}) = \mu_{t,f}.$$

The oracle quantile path is generated by a six-dimensional dynamic linear state with local level, local slope, and two harmonic pairs:

$$\mu_{t,f} = \mathbf{F}^\top \boldsymbol{\theta}_{t,f}, \quad \mathbf{F} = (1, 0, 1, 0, 1, 0)^\top,$$

with local linear trend and two sine–cosine harmonic pairs. The seasonal period is 90, the harmonics are 1 and 2, and the state innovation covariance $\mathbf{W}_\theta$ has diagonal square roots

$$\sqrt{\text{diag}(\mathbf{W}_\theta)} = (0.005, 0.00002, 0.004, 0.004, 0.003, 0.003).$$

The initial covariance is $\mathbf{C}_0 = 0.01\mathbf{I}_6$, and each simulated trajectory is initialized deterministically at the family-specific state in Table 4. For a fixed family, the same latent path $\{\mu_{t,f}\}$ is used across the three values of p ; only the centering constant $H_f^{-1}(p)$ changes.

7.2 Single-Quantile TT500 Fit and Forecast Protocol

Each source trajectory contains $T_{\text{warm}} = 2000$ warmup values and $T_{\text{main}} = 10000$ main source values. All indices below refer to the main source scale after warmup removal. The fitting origin is source index 9000, and the held-out block is 9001:10000. The two fit-window sizes are $T_{\text{fit}} = 500$ and $T_{\text{fit}} = 5000$, corresponding to training windows 8501:9000 and 4001:9000. Forecast evaluation uses a rolling-origin, no-refit state-update protocol over the held-out block. The primary protocol has maximum lead $H_{\text{max}} = 30$ and origin stride 30. The article-facing tables are tied to the verified

Family	Raw error distribution H_f	Initial level	Initial slope	Seasonal settings
Gaussian	$\mathcal{N}(0, 10^2)$	40	0.012	harmonics (24, 0.35), (8, -0.80)
Laplace	Laplace with scale 10	35	0.011	harmonics (28, -0.15), (10, 0.75)
Gaussian mixture	$0.1\mathcal{N}(0, 0.5^2) + 0.9\mathcal{N}(1, 15^2)$	45	0.014	harmonics (32, 0.85), (12, -1.25)

Table 4: TT500 dynamic data-generating mechanisms. Raw errors are shifted by $H_f^{-1}(p)$ so that the p -quantile of the observation error is zero; the same latent dynamic quantile path is then shared across target levels within each family. Seasonal entries report amplitude and phase for the two harmonic pairs in the initial state.

protocol fields, source hashes, and row manifests exported by the validation study.

7.3 Single-Quantile TT500 Competing Methods

For each family and target quantile level in the TT500 validation bundle, the Q-DESN comparison varies the working likelihood, readout prior, and inference method specified by the validation manifests. To keep table labels compact, rows labeled QDESN use the AL working likelihood and rows labeled exQDESN use the exAL working likelihood; the suffix gives the readout prior, either ridge or regularized horseshoe (RHS). The reservoir specification is fixed within this grid and is recorded through the validation provenance, including the reservoir profile, random seed, source hashes, package version, branch, commit, and run tag. Dynamic quantile linear model (DQLM) and extended dynamic quantile linear model (exDQLM) baselines are fit to the same family, quantile level, fit-window, and forecast-origin protocol.

The primary comparison does not use repeated data-generating roots, a broad reservoir hyperparameter search, non-Bayesian ESN readouts, or post hoc multi-quantile synthesis. Those omissions are deliberate: the goal is to test whether Q-DESN readouts can be competitive with structured dynamic quantile linear models under linear dynamic mechanisms that favor the structured baselines.

7.4 Single-Quantile TT500 Criteria for Comparison

Let $\hat{q}_t$ denote a posterior or variational summary of the fitted conditional p -quantile, such as the posterior median of the readout location. Because the oracle path μ_t is known, fit-window recovery can be evaluated using quantile mean absolute error, quantile RMSE, bias, correlation, and check loss at the target level p . Forecast accuracy is evaluated only on outcomes not used for fitting, under common forecast origins and leads for every model. The article-facing tables report fit recovery, forecast scoring, and runtime separately; row-completion and source-hash diagnostics remain in the guarded interface files and generated manifest.

For single-level quantile fits, the primary proper score is pinball loss at the fitted level. Quantile MAE and quantile RMSE compare the fitted or forecast quantile summary directly with the known

oracle quantile path, while pinball loss scores realized observations relative to the reported quantile. These metrics therefore answer different questions and should not be collapsed into one rank. Runtime is reported as a computational criterion for comparing posterior simulation with VB-LD, not as a statistical accuracy measure.

Because the simulation uses one reproducible dynamic root for each family and quantile level, the tables report deterministic criterion values for that source design rather than Monte Carlo means over repeated data sets. Any boldface or ranking in the final tables is interpreted within a fixed family, quantile level, fit-window, model family, and inference comparison. A model row is eligible for interpretation only if the shared interface reports successful completion, valid source hashes, valid fit and forecast windows, nonmissing fit-recovery metrics, nonmissing rolling-origin forecast metrics, and an acceptable health or signoff status.

7.5 TT500 Fit and Forecast Results

Table 5 records the compact protocol for the verified TT500 validation assets. Tables 6–8 report the family-specific TT500 summaries. The entries are generated from lead-level rolling-origin interface rows: fit RMSE measures recovery of the known oracle quantile path on the training window, forecast MAE measures oracle-quantile forecast error averaged across scored rolling-origin lead-target pairs, pinball loss scores realized observations at the fitted quantile level, and runtime is retained in the generated provenance files. Boldface marks the lowest value within each inference panel and quantile level, not a global ranking across different statistical and computational criteria. Some exQDESN RHS cells use audited compact replacements for earlier full-grid exports; the generated manifest records the replacement source, profile, hash, and diagnostic qualification for each such cell. The exQDESN RHS MCMC row uses an audited confirmation run for the VB-selected specifications, with remaining chain-autocorrelation qualifications recorded in the manifest.

Protocol item	Recorded value
Fit size	500 observations
Training window	8501–9000
Forecast block	9001–10000
Forecast origins	every 30 indices from 9000 through 9990
Rolling-origin grid	leads 1–30, stride 30, no refit
Error families	Gaussian, Laplace, Gaussian mixture
Quantile levels	$p = 0.05, 0.25, 0.50$
Inference methods	VB-LD and MCMC
Displayed criteria	Fit RMSE, forecast MAE, check loss

Table 5: Simulation fit-and-forecast protocol used by the manuscript comparison tables. All reported models use the same training window, held-out block, rolling-origin grid, and scoring criteria.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss
<i>VB-LD</i>									
DQLM	2.374	1.445	1.078	2.404	2.513	3.371	1.923	1.109	4.022
exDQLM	2.708	3.126	1.156	2.429	2.022	3.333	1.988	1.125	4.023
QDES ridge	4.665	10.05	1.533	3.949	28.53	8.890	3.254	27.88	14.14
exQDES ridge	4.584	14.68	1.566	3.491	29.61	9.120	3.244	27.59	13.99
QDES RHS	3.561	1.714	1.063	2.821	1.383	3.259	2.463	2.229	4.104
exQDES RHS	3.594	2.491	1.081	2.987	1.715	3.287	2.696	2.005	4.079
<i>MCMC</i>									
DQLM	3.036	3.829	1.103	3.811	2.209	3.328	2.590	1.161	4.029
exDQLM	1.935	16.23	4.313	2.344	6.634	4.132	2.808	1.637	4.109
QDES ridge	4.836	11.04	2.461	3.839	26.11	8.263	3.258	27.75	14.07
exQDES ridge	3.798	16.60	1.652	3.367	27.05	8.496	3.259	27.66	14.03
QDES RHS	3.217	7.930	1.252	2.746	3.310	3.396	1.972	3.791	4.339
exQDES RHS	2.788	2.652	1.076	2.131	3.557	3.437	1.927	3.475	4.285

Table 6: Final TT500 fit-and-forecast comparison for the Gaussian simulation family. Rows compare DQLM, exDQLM, and QDES/exQDES variants; panels separate VB-LD and MCMC. Forecast entries average scored rolling-origin lead-target pairs over leads 1–30 with origin stride 30. Lower values are better for all displayed metrics, and boldface marks the lowest value within each inference panel and quantile level.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss
<i>VB-LD</i>									
DQLM	4.591	3.644	1.868	2.125	3.123	4.485	1.569	1.285	5.078
exDQLM	8.498	9.354	2.146	2.145	2.347	4.427	1.571	1.285	5.077
QDES ridge	9.762	12.00	2.508	4.033	9.353	5.583	2.933	13.51	8.889
exQDES ridge	9.367	9.046	2.099	3.106	12.41	6.088	2.929	13.56	8.908
QDES RHS	7.345	3.871	1.910	2.829	1.803	4.411	1.701	0.9606	5.047
exQDES RHS	8.349	3.038	1.879	2.561	1.623	4.425	2.108	2.663	5.266
<i>MCMC</i>									
DQLM	3.663	9.791	1.968	2.850	3.520	4.548	1.774	1.337	5.082
exDQLM	5.677	22.44	5.154	1.710	6.811	5.106	1.774	2.013	5.206
QDES ridge	9.972	19.77	2.483	4.018	9.954	5.686	2.967	13.41	8.863
exQDES ridge	7.853	9.678	2.063	3.265	10.93	5.821	2.949	12.43	8.453
QDES RHS	6.783	12.14	2.103	2.054	2.474	4.493	1.217	2.765	5.254
exQDES RHS	6.671	2.143	1.859	1.564	3.921	4.608	1.263	3.248	5.334

Table 7: Final TT500 fit-and-forecast comparison for the Laplace simulation family. Rows compare DQLM, exDQLM, and QDES/exQDES variants; panels separate VB-LD and MCMC. Forecast entries average scored rolling-origin lead-target pairs over leads 1–30 with origin stride 30. Lower values are better for all displayed metrics, and boldface marks the lowest value within each inference panel and quantile level.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss
<i>VB-LD</i>									
DQLM	2.725	5.169	1.610	1.958	4.675	4.795	1.598	1.840	5.541
exDQLM	4.615	9.234	1.895	1.834	3.976	4.711	1.595	1.859	5.542
QDESN ridge	8.485	10.27	2.407	5.038	15.93	6.745	4.385	8.481	7.131
exQDESN ridge	8.246	9.650	2.286	4.262	14.31	6.386	4.411	8.758	7.183
QDESN RHS	6.766	3.516	1.572	3.463	1.717	4.491	2.102	1.363	5.445
exQDESN RHS	5.779	3.220	1.567	3.173	2.346	4.506	2.669	1.387	5.443
<i>MCMC</i>									
DQLM	2.623	3.003	1.505	4.740	2.487	4.540	2.226	1.932	5.556
exDQLM	2.554	23.91	5.857	1.381	8.941	5.649	2.408	2.207	5.607
QDESN ridge	8.180	8.722	1.930	5.094	16.82	6.924	4.273	8.097	7.012
exQDESN ridge	6.316	4.912	1.619	3.917	13.17	6.236	4.557	7.532	6.780
QDESN RHS	6.269	4.440	1.562	2.659	4.246	4.690	1.229	3.141	5.690
exQDESN RHS	4.176	2.028	1.510	2.069	5.376	4.803	1.384	3.423	5.727

Table 8: Final TT500 fit-and-forecast comparison for the Gaussian mixture simulation family. Rows compare DQLM, exDQLM, and QDESN/exQDESN variants; panels separate VB-LD and MCMC. Forecast entries average scored rolling-origin lead-target pairs over leads 1–30 with origin stride 30. Lower values are better for all displayed metrics, and boldface marks the lowest value within each inference panel and quantile level.

7.6 TT500 Reproducibility and Limitations

The TT500 result tables are generated by the tracked final-validation config and table-builder script. The generated manifest records the full validation paths, interface hashes, source-registry hash, table hashes, and row counts; the Article-Q-DESN commit containing the manifest provides the article-side provenance. The tables should therefore be interpreted as a final TT500 article-facing validation bundle, not as a claim about TT5000 MCMC, TT100 alternatives, multiple dynamic roots, or an unreported reservoir hyperparameter search. The exQDESN RHS MCMC entries are artifact-complete confirmation fits for the VB-selected specifications; two cells retain high-autocorrelation qualifications in the generated manifest.

7.7 Joint Multi-Quantile Synthetic Validation

The second simulation study evaluates whether a shared quantile-vector readout can recover several conditional quantile paths simultaneously and forecast a held-out dynamic block under known data-generating mechanisms. This design is not a replacement for the TT500 comparison above. It targets a different question: whether joint Q-DESN readouts produce coherent multi-quantile summaries when the oracle conditional quantiles are available.

The four VB rows vary the likelihood and readout structure while keeping the RHS prior fixed. Rows labeled QDESN use the AL working likelihood, and rows labeled exQDESN use the exAL working likelihood. The modifier “joint” denotes one quantile-vector readout fit across

$$\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}.$$

The modifier “independent” denotes separately fit single-quantile readouts assembled over the same

grid and then passed through the same monotone rearrangement before scoring. This convention separates the likelihood choice from the readout structure and avoids treating implementation labels as model names.

The validation uses nine synthetic data-generating mechanisms, a fixed reservoir design, a 500-observation fit window, and a 1000-observation held-out block. The three bridge mechanisms align with the Gaussian, Laplace, and Gaussian-mixture families in the single-quantile study. The six stress mechanisms introduce Student- t errors, asymmetric tails, heteroskedastic seasonality, persistent heavy tails, regime shifts, and nonlinear reservoir-friendly dynamics. Forecasts use the no-refit origin protocol summarized in Table 9. The VB rows compare fit-window recovery and held-out forecast behavior. Quantile summaries are compared with the known conditional quantiles, and realized observations are scored through check loss, grid CRPS, hit-rate diagnostics, and interval diagnostics. Scores use the post-rearrangement monotone quantile grid; pre-rearrangement crossings are retained as numerical stability diagnostics.

Item	Value
Synthetic mechanisms	Three bridge mechanisms and six stress mechanisms with known conditional quantiles.
Quantile grid	0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95
Fit window	500 observations after DESN washout.
Held-out block	1000 observations reserved for held-out validation.
VB comparison rows	Joint and independent QDESN and exQDESN readouts, all with the RHS prior.
Forecast design	No-refit rolling-origin forecasts over a common held-out origin grid.
Reported quantiles	Scores use the post-rearrangement monotone quantile grid; pre-rearrangement crossings are retained as diagnostics.
MCMC reference	VB-initialized MCMC fit-window reference for the selected joint QDESN RHS row.

Table 9: Joint multi-quantile validation protocol. The table reports the statistical design used for the article-facing summaries; source hashes and run-level provenance are recorded in the generated manifest.

Model	Fit MAE	Forecast MAE	Check loss	Grid CRPS	Hit-rate error	Raw crossings
Joint QDESN RHS	0.089	0.096	0.161	0.368	0.018	2
Independent QDESN RHS	0.096	0.098	0.161	0.368	0.016	71
Joint exQDESN RHS	0.130	0.145	0.163	0.370	0.036	0
Independent exQDESN RHS	0.125	0.133	0.162	0.369	0.033	0

Table 10: Compact joint multi-quantile synthetic validation summary averaged over the nine mechanisms in Table 9; this is not a single empirical data set. Model names identify the readout structure and AL/exAL likelihood. Fit and forecast MAE are computed against known conditional quantiles, check loss scores realized observations, grid CRPS summarizes the reported monotone quantile grid, hit-rate error summarizes marginal calibration, and raw crossings count pre-rearrangement numerical crossings.

Scenario	Joint QDES	Independent QDES	Joint exQDES	Independent exQDES
Asymmetric Laplace Tail	0.089 (0.132)	0.087 (0.141)	0.144 (0.209)	0.105 (0.168)
Gaussian-Mixture Bridge	0.097 (0.128)	0.101 (0.136)	0.136 (0.162)	0.131 (0.162)
Heteroskedastic Seasonal	0.088 (0.118)	0.093 (0.128)	0.093 (0.122)	0.086 (0.114)
Laplace Bridge	0.085 (0.122)	0.101 (0.140)	0.139 (0.183)	0.143 (0.188)
Nonlinear Reservoir Friendly	0.102 (0.144)	0.121 (0.159)	0.200 (0.249)	0.198 (0.248)
Normal Bridge	0.090 (0.114)	0.085 (0.113)	0.146 (0.164)	0.123 (0.147)
Persistent Heavy Tail	0.126 (0.164)	0.129 (0.171)	0.137 (0.175)	0.136 (0.177)
Regime Shift	0.120 (0.137)	0.096 (0.125)	0.176 (0.220)	0.142 (0.183)
Student-t Location-Scale	0.068 (0.089)	0.073 (0.101)	0.134 (0.165)	0.130 (0.161)

Table 11: Scenario-level held-out forecast distance to true conditional quantiles for the same joint multi-quantile synthetic validation suite. Entries report forecast MAE with RMSE in parentheses for each VB row; lower values indicate closer agreement with the oracle conditional quantile paths, and boldface marks the lowest displayed MAE within each scenario.

Table 10 reports that, after averaging over the nine synthetic mechanisms rather than over a single empirical data set, the joint QDES RHS row has the lowest average held-out MAE to the known conditional quantiles among the four VB rows. Its forecast MAE is 0.096, compared with 0.098 for the independent QDES comparator, 0.145 for joint exQDES, and 0.133 for independent exQDES. The margin between the joint and independent AL rows is small, so the main interpretation is not a broad dominance claim. Rather, the joint readout is competitive in forecast distance while requiring far fewer pre-rearrangement crossing corrections than the independent AL row. The exQDES rows have fewer raw crossings in this validation bundle, but they are farther from the oracle conditional quantiles. Table 11 gives the same held-out forecast-distance comparison by synthetic mechanism, replacing the heat-map view with numerical entries that can be checked directly.

The VB-initialized MCMC reference run is restricted to the fit window for the selected joint QDES RHS specification. It is not a rolling-origin forecast validation. Instead, it checks whether VB provides a stable initializer for posterior simulation and whether the fitted quantile paths remain close to the known oracle paths. Across the nine scenarios, the MCMC reference run produces finite draws, noncrossing raw and reported quantile grids, and pass-level scenario diagnostics. This result supports the selected VB fit as an MCMC initializer for the primary joint QDES RHS row, while leaving held-out forecast comparison to the VB validation layer.

The generated provenance tables retain the full scenario summaries, MCMC scenario diagnostics, source-hash checks, and run-level diagnostic status. The MCMC distance plots are retained as generated diagnostics but are not printed in the main text because they are fit-reference checks rather than held-out forecast comparisons. Those records are not additional statistical claims; they document that the article tables and figures were generated from verified validation outputs without rerunning inference. The joint validation should therefore be interpreted as fixed-design evidence for coherent multi-quantile reporting under the specified reservoir, scenarios, quantile grid, and forecast protocol.

8 Application: GloFAS Streamflow Forecast Calibration

The empirical application considers medium-range streamflow forecast calibration using the Global Flood Awareness System (GloFAS) and a reference gauge record. GloFAS is a global hydrological forecast and monitoring system jointly developed by the European Commission and the European Centre for Medium-Range Weather Forecasts (ECMWF) and operated within the Copernicus Emergency Management Service (Alfieri et al., 2013; Copernicus Emergency Management Service, 2026). Its operational forecast and reforecast products provide ensemble river-discharge information for flood early warning and forecast verification (Harrigan et al., 2023). Previous calibration work has focused on hydrological model parameters using daily streamflow observations (Hirpa et al., 2018); here Q-DESN is used as a statistical quantile-calibration layer for one forecast system and one audited forecast origin.

8.1 Data and Forecast-Origin Protocol

Let T denote a forecast origin and let H be the issued forecast horizon. The historical reference record is y_t , the transformed USGS streamflow through time T . The retrospective GloFAS product over the same period is g_t^{ret} , and $g_{T+h,m}^{\text{ens}}$ denotes GloFAS ensemble member m issued at origin T for horizon $h = 1, \dots, H$. The values $y_{T+1}, \dots, y_{T+H}$ are not available at the forecast origin. They are held out during prediction and used only later for validation.

The selected manuscript-facing run uses the forecast origin 2022-12-25 and scores seven fitted quantile levels over 28 issued target dates. The application is therefore an audited reference case rather than a basin-wide or multi-origin operational evaluation. This restriction is intentional: it keeps the data extraction, forecast-origin alignment, posterior prediction contract, and promotion manifest inspectable before broader comparisons are reported.

8.2 Application Model

For $t \leq T$, the application uses two deterministic Q-DESN feature maps. The reference block $\mathbf{x}_t^{\text{ref}}$ is driven by the transformed reference history and origin-available covariates. The discrepancy block $\mathbf{x}_t^{\text{disc}}$ is driven by the retrospective discrepancy $g_t^{\text{ret}} - y_t$, using the same audited preprocessing, lag, reservoir, reducer, and washout conventions. For $t > T$, both feature maps are recomputed inside posterior prediction, because output lags may depend on the latent future reference path. For each posterior draw, the forecast-window discrepancy reservoir is driven by the horizon-keyed contrast between the empirical GloFAS ensemble quantile $\hat{q}_{p_0, T+h}^{\text{ens}}$ and the draw-specific future reference path. Thus forecast-window design rows are posterior-predictive quantities, not fixed rows that can be constructed before prediction.

For a fixed quantile level p_0 , the calibrated reference quantile is represented by one Q-DESN readout and the GloFAS discrepancy by a second readout,

$$q_{p_0,t}^{\text{ref}} = \mathbf{x}_t^{\text{ref}\top} \boldsymbol{\beta}_{p_0}, \quad d_{p_0,t}^{\text{glo}} = \mathbf{x}_t^{\text{disc}\top} \boldsymbol{\alpha}_{p_0}, \quad q_{p_0,t}^{\text{glo}} = q_{p_0,t}^{\text{ref}} + d_{p_0,t}^{\text{glo}}.$$

The path $q_{p_0,t}^{\text{ref}}$ is the target reference-process quantile, while $d_{p_0,t}^{\text{glo}}$ is the quantile-specific discrepancy between GloFAS and the reference process. The retrospective and ensemble GloFAS products inform the forecast-system quantile path $q_{p_0,t}^{\text{glo}}$; the calibrated forecast target remains $q_{p_0,T+h}^{\text{ref}}$.

At forecast origin T , the issued GloFAS ensemble supplies information about the forecast-system quantile at each covered target date, not a direct forecast of the reference quantile. The Bayesian prediction object is draw-based. For posterior draw $s = 1, \dots, S$, the draw contains a future reference path, the implied forecast-window reservoir states, and the readout coefficients:

$$q_{p_0,T+h}^{\text{ref},(s)} = \mathbf{x}_{T+h}^{\text{ref},(s)\top} \boldsymbol{\beta}_{p_0}^{(s)}, \quad q_{p_0,T+h}^{\text{glo},(s)} = q_{p_0,T+h}^{\text{ref},(s)} + \mathbf{x}_{T+h}^{\text{disc},(s)\top} \boldsymbol{\alpha}_{p_0}^{(s)}.$$

Equivalently, $q_{p_0,T+h}^{\text{ref},(s)} = q_{p_0,T+h}^{\text{glo},(s)} - d_{p_0,T+h}^{\text{glo},(s)}$. Posterior means, medians, intervals, and monotone multi-quantile summaries are computed only after this draw-level construction.

The working-likelihood specification is

$$\begin{aligned} y_t &| q_{p_0,t}^{\text{ref}}, \sigma_{\text{ref}}, \gamma_{\text{ref}} \sim \text{exAL}_{p_0}(q_{p_0,t}^{\text{ref}}, \sigma_{\text{ref}}, \gamma_{\text{ref}}), \\ g_t^{\text{ret}} &| q_{p_0,t}^{\text{glo}}, \sigma_{\text{glo}}, \gamma_{\text{glo}} \sim \text{exAL}_{p_0}(q_{p_0,t}^{\text{glo}}, \sigma_{\text{glo}}, \gamma_{\text{glo}}), \\ g_{T+h,m}^{\text{ens}} &| q_{p_0,T+h}^{\text{glo}}, \sigma_{\text{glo}}, \gamma_{\text{glo}} \sim \text{exAL}_{p_0}(q_{p_0,T+h}^{\text{glo}}, \sigma_{\text{glo}}, \gamma_{\text{glo}}), \quad h = 1, \dots, H, \quad m = 1, \dots, M. \end{aligned}$$

The issued ensemble members enter as conditionally independent likelihood contributions given the forecast-system quantile path and source-specific likelihood parameters. ECMWF ensemble forecasts use perturbed initial states and model perturbations to produce forecast alternatives (European Centre for Medium-Range Weather Forecasts, 2026); in the Q-DESN calibration layer, this supports treating GloFAS members at a common forecast origin and horizon as exchangeable draws around $q_{p_0,T+h}^{\text{glo}}$. The default application model shares σ_{glo} and γ_{glo} between retrospective and issued-ensemble GloFAS rows.

For this reference-case analysis, the seven fitted quantile levels are 0.05, 0.15, 0.35, 0.50, 0.65, 0.80, and 0.95. They are fitted independently with the AL special case and the VB approximation. This application therefore uses the independent-fit and synthesis workflow, not the joint quantile-vector readout of Section 3.2. The draw-level predictive summaries are then passed through the monotone synthesis step. The selected manuscript output additionally uses a no-refit spread calibration at the synthesis stage, centered at quantile 0.50, with multiplicative factor 1.4 and additive half-width 0.5. This adjustment changes only the synthesized forecast-window bands; it does not alter the reservoir features, likelihood, priors, or fitted readout coefficients.

The selected fit uses separate DESN feature maps for the reference quantile path and the GloFAS discrepancy path. Both maps have depth $D = 4$, layer widths (100, 100, 100, 100), reducer widths (100, 100, 100), memory 300, washout 500, spectral radii (0.95, 0.95, 0.95, 0.95), recurrent sparsity (0.03, 0.03, 0.03, 0.03), and dense input matrices. The reference map uses leak rates (0.0225, 0.0225, 0.0225, 0.0225) and global/bias input scales 0.16/0.16, and reservoir seed 20260512; the discrepancy map uses leak rates (0.0075, 0.0075, 0.0075, 0.0075), global/bias input scales 0.28/0.28, and reservoir seed 20261521. The two readout blocks use separate regularized-horseshoe global

scales, $\tau_{0,\text{ref}} = 0.001$ and $\tau_{0,\text{disc}} = 0.03$, respectively. The tracked config snapshot and selection manifest record the engine commit, source hashes, selected file hashes, and current manuscript aliases for this run.

8.3 Reference Application Results

Table 12 reports the forecast scores for the selected reference run. Relative to raw GloFAS, the Q-DESN calibration reduces the mean check loss from 0.7639 to 0.3339, the interval score from 13.0538 to 4.1402, and the quantile-grid CRPS from 1.4424 to 0.6859. Mean empirical interval coverage increases from 0.000 to 0.833. These summaries describe one audited forecast origin and should be read as a reference-case calibration result rather than as a multi-origin operational validation.

Table 12: Forecast scoring for the selected GloFAS application run. Scores are computed on the transformed streamflow scale for the audited forecast origin 2022-12-25. Lower check loss, interval score, and CRPS are better; mean coverage reports empirical interval coverage across the scored nominal intervals.

Model	Horizons	Check	Interval	CRPS	Coverage	Reduction
Q-DESN calibration	28	0.3339	4.1402	0.6859	0.833	56.3%
Raw GloFAS	28	0.7639	13.0538	1.4424	0.000	Reference

Discrepancy-corrected Q-DESN quantile paths after monotone synthesis

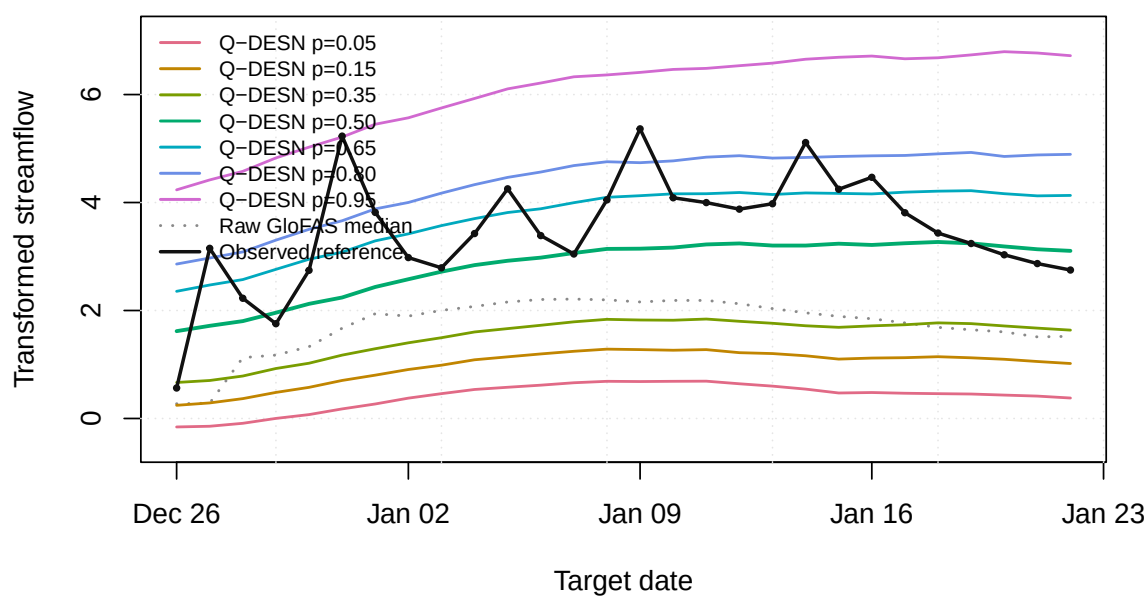


Figure 2: Raw GloFAS quantile paths, discrepancy-corrected Q-DESN quantile paths after monotone synthesis and synthesis-time spread calibration, and held-out reference streamflow over the issued forecast window for the selected GloFAS application run.

Q-DESN Synthesized Predictive Quantile Bands

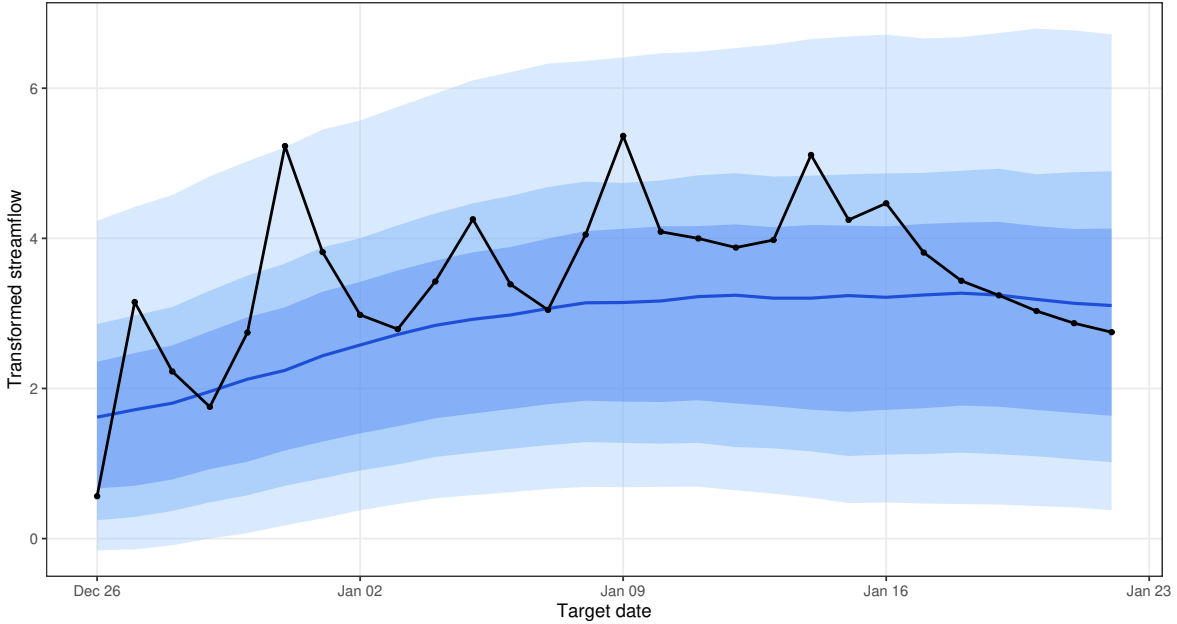


Figure 3: Posterior forecast-window diagnostic for the selected GloFAS application run. The display checks the synthesized and spread-calibrated quantile bands around the forecast origin and the issued target dates used in Table 12.

The selected run passed the source-hash, file-hash, score-table, and VB convergence checks used by the article workflow. The VB fits used median 150; max 150 iterations. Future model specifications can replace this reference case through the same selection manifest and current-output registry, without changing the application model statement.

9 Application: European Electricity Price Forecasting

As a second empirical case, we consider probabilistic day-ahead electricity price forecasting using the PriceFM benchmark of Yu et al. (2026). The released data cover European bidding regions at 15-minute resolution and include day-ahead price, load, solar, and wind forecasts. PriceFM itself is a graph-informed foundation model: it uses region-level exogenous information and a sparse graph mask derived from the European transmission topology. Our article-side comparison is therefore designed as a scoped benchmark against cached fold-aligned PriceFM predictions, not as a claim about the full paper-wide PriceFM evaluation.

9.1 Comparison Protocol

The article-facing benchmark contains 114 region/fold rows, spanning 38 European bidding regions and 3 rolling folds. Each row is evaluated on the same fold-aligned test timestamps used by the cached PriceFM predictions. The scoring metric is original-unit average quantile loss (AQL) on the

PriceFM paper quantile grid 0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90. Throughout this section, Δ AQL denotes Q-DESN AQL minus PriceFM AQL, so negative values favor Q-DESN.

The benchmark panel is assembled from audited validation registries and previously computed paper-quantile score files. This assembly step is registry-only: it does not fit new models and it does not alter the cached PriceFM predictions. Across the benchmark, 58 selected rows use target-only information and 56 selected rows use graph-derived PriceFM neighborhood summaries. The benchmark is interpreted as an application of the selected Q-DESN forecasting workflow recorded by the validation registries. It is not used as evidence for the joint quantile-vector RHS prior unless a future PriceFM run explicitly fits and promotes that joint readout under the same fold-aligned contract.

9.2 Full-Surface Results

Across the aligned benchmark, Q-DESN has lower AQL than PriceFM in 66 of 114 rows (57.9%), is near-tied in 12 rows, and trails PriceFM in 36 rows. The mean Q-DESN AQL is 6.826, compared with mean PriceFM AQL 7.039, giving mean Δ AQL -0.213 and median Δ AQL -0.083. The advantage is therefore modest rather than uniform. Table 13 shows that the result is strongest in fold 2, still favorable in fold 1, and essentially tied in fold 3.

Table 13: PriceFM benchmark summary. AQL is computed on the original electricity-price scale using the PriceFM paper quantile grid. The last two columns report Q-DESN AQL minus PriceFM AQL; negative values favor Q-DESN. Near ties are retained as a separate monitoring category rather than counted as wins.

Scope	Rows	Q-DESN wins	Near ties	PriceFM wins	Q-DESN AQL	PriceFM AQL	Mean Δ	Median Δ
Overall	114	66	12	36	6.826	7.039	-0.213	-0.083
Fold 1	38	22	6	10	7.047	7.297	-0.250	-0.073
Fold 2	38	30	2	6	6.429	6.826	-0.397	-0.295
Fold 3	38	14	4	20	7.001	6.993	0.008	0.281

The input set also matters. Graph-derived covariates give the clearest average advantage, whereas target-only rows are closer to neutral. This comparison should not be read as a causal attribution to graph features, because input sets and selected model specifications vary across region/fold rows. It is a diagnostic summary of where the selected registry is most competitive.

Table 14: PriceFM benchmark by input set. Graph-derived inputs include selected rows using graph-neighborhood summaries or graph-neighborhood lag features. Target-only inputs use only the target-region information available to the local forecasting workflow.

Input set	Rows	Q-DESN wins	Near ties	PriceFM wins	Mean Δ	Median Δ
Graph-derived inputs	56	34	5	17	-0.366	-0.113
Target-only inputs	58	32	7	19	-0.064	-0.046

9.3 Horizon Diagnostics and Remaining Gaps

Horizon-block diagnostics are available for 72 of the 114 rows. These diagnostics preserve the same convention as the aggregate comparison: negative Δ values favor Q-DESN. They should be read as a partial diagnostic layer rather than as a second full-surface registry, because not every audited source artifact retained horizon-block scores.

Table 15: Available horizon-block diagnostics for the PriceFM benchmark. Rows denote scored region/fold and horizon-block combinations for the subset with retained horizon diagnostics. Negative Δ values favor Q-DESN.

Horizon block	Rows	Q-DESN wins	Mean Δ	Median Δ
1-24	72	22	0.421	0.400
25-48	72	45	-0.683	-0.382
49-72	72	34	-0.196	0.032
73-96	72	42	-0.152	-0.114

The resulting evidence supports a conservative scoped statement. Q-DESN is a competitive reservoir baseline against cached fold-aligned PriceFM predictions on the aligned benchmark panel, with better mean and median AQL in aggregate. It is not uniformly better than PriceFM: fold 3 is nearly tied on average, and PriceFM remains better for a nontrivial subset of region/fold rows. The remaining empirical work is therefore targeted model repair for those rows and a fully joint PriceFM run if application-level evidence for the joint multi-quantile readout is desired.

10 Discussion

Q-DESN treats the fixed DESN reservoir as a nonlinear feature map and places Bayesian quantile inference on the readout. This separation is useful computationally: posterior simulation and VB-LD update only the readout, likelihood, and shrinkage parameters, while the recurrent weights, reducers, scaling constants, and washout convention remain fixed. The empirical results therefore evaluate a conditional Bayesian readout workflow. They should not be interpreted as posterior averaging over reservoir architectures or as a fully Bayesian treatment of recurrent network uncertainty.

The likelihood is also deliberately limited in interpretation. The AL and exAL families are used as quantile-targeting working likelihoods. They provide posterior targets and conditionally tractable augmented updates, but credible bands from the fitted working likelihood are not automatic frequentist coverage guarantees under misspecification. The simulation and application sections therefore report fit recovery, forecast scoring, empirical coverage, runtime, convergence, and provenance diagnostics rather than relying on posterior summaries alone. Calibration of learning rates, sandwich adjustments, or general-Bayes scaling remains an important sensitivity direction, especially for composite multi-quantile likelihoods.

The multi-quantile construction makes the same distinction. Independent single-level Q-DESN fits followed by isotonic regression or rearrangement produce monotone reported quantile curves, but they do not define a joint posterior over quantile-indexed readouts. The joint quantile-vector extension shares information across levels through adjacent RHS slope innovations and gives a soft non-crossing mechanism. It does not impose the non-crossing inequalities as hard constraints. When exact monotonicity is required for downstream use, the reported curve should still be obtained through an ordered-intercept parameterization, projection, rearrangement, or another explicitly declared monotone contract.

The validation evidence is correspondingly scoped. The TT500 experiments test single-quantile recovery and held-out forecasting under a pinned dynamic simulation design. The joint multi-quantile synthetic study checks coherence under known conditional quantile paths. The GloFAS application is an audited one-origin streamflow calibration case using independent AL-VB fits and monotone synthesis, while the PriceFM comparison is a fold-aligned benchmark against cached Phase-I predictions. These studies support the reported workflows under their own protocols, not a universal ranking of Q-DESN over all dynamic quantile or foundation-model alternatives.

Several extensions are natural. On the statistical side, future work should study calibrated composite-likelihood scaling, hard or probabilistic non-crossing constraints, richer covariance structures for quantile innovations, and sensitivity to reservoir scaling on bounded feature domains. On the computational side, the joint model calls for sparse precision solvers, selected covariance extraction for VB, and diagnostics comparing VB-LD with MCMC on small joint problems. On the application side, broader multi-origin GloFAS studies and explicitly joint PriceFM runs are needed before the joint quantile-vector prior can be interpreted as application-level evidence rather than a synthetic validation result.

Acknowledgments

References

- Alfieri, L., Burek, P., Dutra, E., Krzeminski, B., Muraro, D., Thielen, J., and Pappenberger, F. (2013). GloFAS – global ensemble streamflow forecasting and flood early warning. *Hydrology and Earth System Sciences*, 17:1161–1175.
- Bai, Y.-T., Jia, W., Jin, X.-B., Su, T.-L., Kong, J.-L., and Shi, Z.-G. (2023). Nonstationary time series prediction based on deep echo state network tuned by bayesian optimization. *Mathematics*, 11(6):1503.
- Barata, R., Prado, R., and Sansó, B. (2022). Fast inference for time-varying quantiles via flexible dynamic models with application to the characterization of atmospheric rivers. *The Annals of Applied Statistics*, 16(1):247 – 271.

- Barlow, R. E. and Brunk, H. D. (1972). The isotonic regression problem and its dual. *Journal of the American Statistical Association*, 67(337):140–147.
- Bissiri, P. G., Holmes, C. C., and Walker, S. G. (2016). A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130.
- Bonas, M., Wikle, C. K., and Castruccio, S. (2024). Calibrated forecasts of quasi-periodic climate processes with deep echo state networks and penalized quantile regression. *Environmetrics*, 35(1):e2833.
- Bondell, H. D., Reich, B. J., and Wang, H. (2010). Noncrossing quantile regression curve estimation. *Biometrika*, 97(4):825–838.
- Cardoso, F. C., Berri, R. A., Borges, E. N., Dalmazo, B. L., Lucca, G., and Dias de Mattos, V. L. (2024). Echo state network and classical statistical techniques for time series forecasting: A review. *Knowledge-Based Systems*, 293:111639.
- Carvalho, C. M., Polson, N. G., and Scott, J. G. (2010). The horseshoe estimator for sparse signals. *Biometrika*, 97(2):465–480.
- Chernozhukov, V., Fernández-Val, I., and Galichon, A. (2010). Quantile and probability curves without crossing. *Econometrica*, 78(3):1093–1125.
- Copernicus Emergency Management Service (2026). Global flood awareness system documentation. <https://confluence.ecmwf.int/display/CEMS/Global+Flood+Awareness+System>. Accessed 11 May 2026.
- Ehlers, P. J., Nurdin, H. I., and Soh, D. (2025). Improving the performance of echo state networks through state feedback.
- European Centre for Medium-Range Weather Forecasts (2026). Forecast user guide: Section 5.1 generation of the ensemble. <https://confluence.ecmwf.int/display/FUG/Section+5.1+Generation+of+the+Ensemble>. Accessed 11 May 2026.
- Fan, H.-T., Wang, W., and Jin, Z. (2017). Performance optimization of echo state networks through principal neuron reinforcement. In *2017 International Joint Conference on Neural Networks (IJCNN)*, pages 1717–1723.
- Gallicchio, C., Micheli, A., and Pedrelli, L. (2018). Design of deep echo state networks. *Neural Networks*, 108:33–47.
- Gandhi, M., Lee, K., Pan, Y., and Theodorou, E. (2018). Propagating uncertainty through the tanh function with application to reservoir computing.
- Gneiting, T. (2011). Quantiles as optimal point forecasts. *International Journal of Forecasting*, 27(2):197–207.

- Gneiting, T. and Katzfuss, M. (2014). Probabilistic forecasting. *Annual Review of Statistics and Its Application*, 1:125–151.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Gonçalves, K. C. M., Migon, H. S., and Bastos, L. S. (2020). Dynamic quantile linear models: A Bayesian approach. *Bayesian Analysis*, 15(2):335–362.
- Guerra, M., Scardapane, S., and Bianchi, F. M. (2023). Probabilistic load forecasting with reservoir computing. *IEEE Access*, 11:145989–146002.
- Harrigan, S., Zsótér, E., Cloke, H., Salamon, P., and Prudhomme, C. (2023). Daily ensemble river discharge reforecasts and real-time forecasts from the operational global flood awareness system. *Hydrology and Earth System Sciences*, 27:1–19.
- Hirpa, F. A., Salamon, P., Beck, H. E., Lorini, V., Alfieri, L., Zsótér, E., and Dadson, S. J. (2018). Calibration of the global flood awareness system (GloFAS) using daily streamflow data. *Journal of Hydrology*, 566:595–606.
- Jaeger, H. (2001). The “echo state” approach to analysing and training recurrent neural networks. Technical Report 148, GMD – German National Research Center for Information Technology, Sankt Augustin, Germany.
- Jaeger, H., Lukoševičius, M., Popovici, D., and Siewert, U. (2007). Optimization and applications of echo state networks with leaky-integrator neurons. *Neural Networks*, 20(3):335–352.
- Ji, F., Lee, J., and Rabe-Hesketh, S. (2025). Valid standard errors for Bayesian quantile regression with clustered and independent data. *Journal of Educational and Behavioral Statistics*. OnlineFirst.
- Jiang, L., Bondell, H. D., and Wang, H. J. (2014). Interquantile shrinkage and variable selection in quantile regression. *Computational Statistics & Data Analysis*, 69:208–219.
- Kim, T. and King, B. R. (2020). Time series prediction using deep echo state networks. *Neural Computing and Applications*, 32(23):17769–17787.
- Koenker, R. (2005). *Quantile Regression*. Cambridge University Press, Cambridge.
- Koenker, R. and Bassett, Jr., G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.
- Kohns, D. and Szendrei, T. (2025). Joint quantile shrinkage: A state-space approach toward non-crossing bayesian quantile models.
- Kozumi, H. and Kobayashi, G. (2011). Gibbs sampling methods for Bayesian quantile regression. *Journal of Statistical Computation and Simulation*, 81(11):1565–1578.

- Lukoševičius, M. (2012). A practical guide to applying echo state networks. In Montavon, G., Orr, G. B., and Müller, K.-R., editors, *Neural Networks: Tricks of the Trade*, pages 659–686. Springer Berlin Heidelberg, Berlin, Heidelberg, second edition.
- Lukoševičius, M. and Jaeger, H. (2009). Reservoir computing approaches to recurrent neural network training. *Computer Science Review*, 3(3):127–149.
- Ly, Z., Zhao, J., Liu, Y., and Wang, W. (2016). Use of a quantile regression based echo state network ensemble for construction of prediction intervals of gas flow in a blast furnace. *Control Engineering Practice*, 46:94–104.
- Maat, J. R., Gianniotis, N., and Protopapas, P. (2018). Efficient optimization of echo state networks for time series datasets. In *2018 International Joint Conference on Neural Networks (IJCNN)*, pages 1–7. IEEE.
- Makalic, E. and Schmidt, D. F. (2016). A simple sampler for the horseshoe estimator. *IEEE Signal Processing Letters*, 23(1):179–182.
- McDermott, P. L. and Wikle, C. K. (2019). Deep echo state networks with uncertainty quantification for spatio-temporal forecasting. *Environmetrics*, 30(3):e2553.
- Nishimura, A. and Suchard, M. A. (2023). Shrinkage with shrunken shoulders: Gibbs sampling shrinkage model posteriors with guaranteed convergence rates. *Bayesian Analysis*, 18(4):1163–1194.
- Piironen, J. and Vehtari, A. (2017). Sparsity information and regularization in the horseshoe and other shrinkage priors. *Electronic Journal of Statistics*, 11(2):5018–5051.
- Saadat, J., Farshad, M., Eliasi, H., and Shokoohi Mehr, K. (2025). Establishing echo state network in order to be used in online application. *Operations Research Forum*, 6(115).
- Singh, P. and Raman, B. (2025). Echo state networks as state-space models: A systems perspective.
- Sriram, K., Ramamoorthi, R. V., and Ghosh, P. (2013). Posterior consistency of Bayesian quantile regression based on the misspecified asymmetric laplace density. *Bayesian Analysis*, 8(2):479–504.
- Sun, C., Song, M., Cai, D., Zhang, B., Hong, S., and Li, H. (2024). A systematic review of echo state networks from design to application. *IEEE Transactions on Artificial Intelligence*, 5(1):23–37.
- Szendrei, T., Bhattacharjee, A., and Schaffer, M. E. (2025). Fused LASSO as non-crossing quantile regression.
- Viehweg, J., Teutsch, P., and Mäder, P. (2025). A systematic study of echo state networks topologies for chaotic time series prediction. *Neurocomputing*, 618:129032.
- Wang, C. and Blei, D. M. (2013). Variational inference in non-conjugate models. *Journal of Machine Learning Research*, 14:1005–1031.

- Wang, J. and Cai, Z. (2024). A composite bayesian approach for quantile curve fitting with non-crossing constraints. *Communications in Statistics – Theory and Methods*.
- Wu, Q., Fokoué, E., and Kudithipudi, D. (2018). On the statistical challenges of echo state networks and some potential remedies.
- Wu, T. and Narisetty, N. N. (2021). Bayesian multiple quantile regression for linear models using a score likelihood. *Bayesian Analysis*, 16(4):1261–1284.
- Wu, Y. and Liu, Y. (2009). Stepwise multiple quantile regression estimation using non-crossing constraints. *Statistics and Its Interface*, 2(3):299–310.
- Yan, M., Huang, C., Bienstman, P., Tiño, P., Lin, W., and Sun, J. (2024). Emerging opportunities and challenges for the future of reservoir computing. *Nature Communications*, 15:2056.
- Yan, Y., Zheng, X., and Kottas, A. (2025). A new family of error distributions for Bayesian quantile regression. *Bayesian Analysis*, pages 1–29. Advance Publication.
- Yang, Y. and Tokdar, S. T. (2017). Joint estimation of quantile planes over arbitrary predictor spaces. *Journal of the American Statistical Association*, 112(519):1107–1120.
- Yang, Y., Wang, H. J., and He, X. (2016). Posterior inference in Bayesian quantile regression with asymmetric laplace likelihood. *International Statistical Review*, 84(3):327–344.
- Yildiz, I. B., Jaeger, H., and Kiebel, S. J. (2012). Re-visiting the echo state property. *Neural Networks*, 35:1–9.
- Yu, K. and Moyeed, R. A. (2001). Bayesian quantile regression. *Statistics & Probability Letters*, 54(4):437–447.
- Yu, R., Gu, C., Stiasny, J., Wen, Q., Dilov, W. S., Qi, L., and Cremer, J. L. (2026). PriceFM: Foundation model for probabilistic electricity price forecasting.
- Zhang, H. and Vargas, D. V. (2023). A survey on reservoir computing and its interdisciplinary applications beyond traditional machine learning. *IEEE Access*, 11:81033–81070.
- Zou, H. and Yuan, M. (2008). Composite quantile regression and the oracle model selection theory. *The Annals of Statistics*, 36(3):1108–1126.